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# Efficient Approaches to Solve Plasma Dispersion Relations with Arbitrary Distributions

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# Contents



Goal of efficiency:

- Fast
- Accurate
- All solutions (without initial guess for root finding)
- Support damped mode
- Arbitrary distributions

Note: There are series other relevant works not mentioned in this presentation, but listed in the references of the above papers.

# 1. Fast and accurate calculate of plasma dispersion function Z

## 1.1 Maxwellian distribution (Xie2024AIP)

$$Z(s) = \frac{1}{\sqrt{\pi}} \int_{-\infty}^{\infty} \frac{e^{-t^2}}{t-s} dt,$$

$$Z(s) = [Z(s^*)]^* + 2i\sqrt{\pi} \exp[-s^2] \quad (y<0).$$

The following is the two-side Taylor expansion approximation:

$$Z(s) \simeq \begin{cases} \sum_{k=0}^{\infty} a_k s^k \simeq i\sqrt{\pi} e^{-s^2} - s\sum_{n=0}^{\infty} (-s^2)^n \frac{\Gamma(1/2)}{\Gamma(n+3/2)}, & s \to 0, \\ \sum_{k=0}^{\infty} a_{-k} s^{-k} \simeq i\sigma\sqrt{\pi} e^{-s^2} - \sum_{n=0}^{\infty} \frac{\Gamma(n+1/2)}{\Gamma(1/2) s^{2n+1}}, & s \to \infty, \end{cases}$$

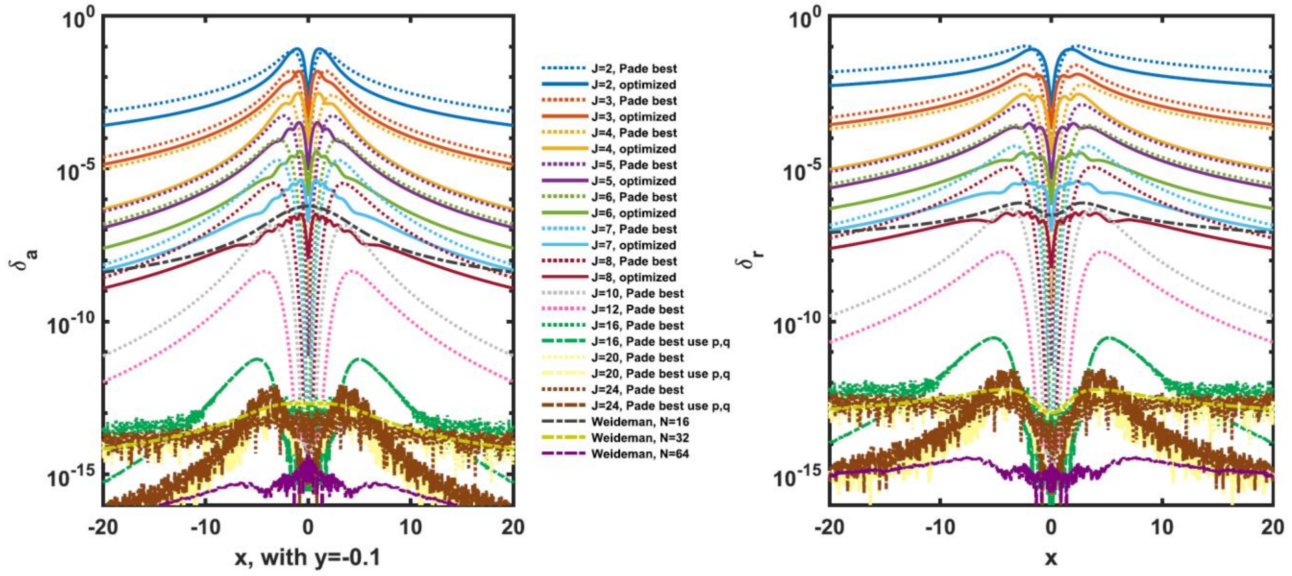



**FIG. 4.** Errors of $Z$ using different $J$ poles coefficients and Weideman coefficients, with $y = -0.1$.

Python code for Z fun with optimized J=8 pole

```
import numpy as np

def zfunJ8(z):
   Zeta = np.zeros_like(z, dtype=complex)
   bj = np.array([0.00383968430671409 - 0.0119854387180615j,
            -0.321597857664957 - 0.218883985607935j,
            2.55515264319988 + 0.613958600684469j,
            -2.73739446984183 + 5.69007914897806j])
   cj = np.array([2.51506776338386 - 1.60713668042405j,
            -1.68985621846204 - 1.66471695485661j,
            0.981465428659098 - 1.70017951305004j,
            -0.322078795578047 - 1.71891780447016j])
   bj = np.concatenate((bj, np.conj(bj[::-1])))
   cj = np.concatenate((cj, -np.conj(cj[::-1])))

   idx = np.imag(z) >= 0
   Zeta[~idx] = 2j * np.sqrt(np.pi) * np.exp(-(z[~idx])**2)
   for j in range(len(bj)):
      Zeta[idx] += bj[j] / (z[idx] - cj[j])
      Zeta[~idx] += np.conj(bj[j] / (np.conj(z[~idx]) - cj[j]))

   return Zeta
```

Pade expansion -> J-pole

$$Z(s) \simeq Z_A^J(s) = \frac{\sum_{l=0}^{J-1} p_l s^l}{q_0 + \sum_{k=1}^{J} q_k s^k} = \sum_{j=1}^{J} \frac{b_j}{s-c_j},$$

Simple code yields high accurate result, J=24, error to 1e-12

NRL formula: J=2 (inaccurate)

Feature: Fast, accurate, rational form (useful, later)

AIP Advances 14, 075007 (2024)

https://github.com/hsxie/gpdf

# 1. Fast and accurate calculate of plasma dispersion function Z

## 1.2 Arbitrary distribution (Xie2013PoP)

$$Z(\zeta) = Z(\zeta, F) = \int_C \frac{F}{z-\zeta} dz,$$

Assuming an expansion,

$$[W(v)]^{-1}F(v) = \sum_{n=-\infty}^{\infty} a_n \rho_n(v), \qquad v \in \mathcal{R}, \tag{18}$$

where $\{\rho_n(v)\}$ is an orthogonal basis set with weight function $W(v)$, i.e.,

$$\int_{-\infty}^{\infty} W(v)\rho_n(v)\rho_m^*(v)dv = A\delta_{m,n}, \tag{19}$$

where the asterisk denotes complex conjugation and $\delta_{m,n}$ is the Kronechker delta. The coefficients are

$$a_n = \frac{1}{A}\int_{-\infty}^{\infty} F(v)\rho_n^*(v)dv. \tag{20}$$

Then,

$$\frac{F(v)}{v-z} = \sum_{n=-\infty}^{\infty} a_n \left[ W(v)\frac{\rho_n(v)}{v-z} \right]. \tag{21}$$

For the upper half plane, we use weight function $W(v) = 1/(L^2+v^2)$ (Ref. 9) and basis functions,

$$\rho_n(v) = \frac{(L+iv)^n}{(L-iv)^n}, \tag{22}$$

which is a Fourier form because $e^{i\theta} = (L+iv)/(L-iv)$ with $v = L\tan(\theta/2)$ and $dv/d\theta = (L^2+v^2)/(2L)$, then $a_n$ can be evaluated using FFT and we can obtain $A = \pi/L$ using $\int_{-\pi}^{\pi} e^{in\theta}e^{-im\theta}d\theta = 2\pi\delta_{m,n}$.

Using residues, for $\Im(\zeta) > 0$, we find the integrals

$$\int_{-\infty}^{\infty} \frac{W(v)}{v-z}\frac{(L+iv)^n}{(L-iv)^n}dv = \begin{cases} \frac{i\pi}{L}\frac{1}{(L-iz)}, & n=0, \\ \frac{2i\pi}{L^2+z^2}\frac{(L+iz)^n}{(L-iz)^n}, & n>0, \\ 0, & n<0. \end{cases} \tag{23}$$

We obtain

$$g^+(z) = \frac{2i}{L^2+z^2}\sum_{n=1}^{\infty} a_n \left(\frac{L+iz}{L-iz}\right)^n + \frac{ia_0}{L(L-iz)}, \qquad \Im(z) > 0. \tag{24}$$

For the lower half plane, $\Im(\zeta) < 0$, we use $F(z) = [F(z^*)]^*$ and obtain

$$Z(\zeta) = [Z(\zeta^*)]^* + 2i\pi f(\zeta), \qquad \Im(\zeta) < 0. \tag{25}$$

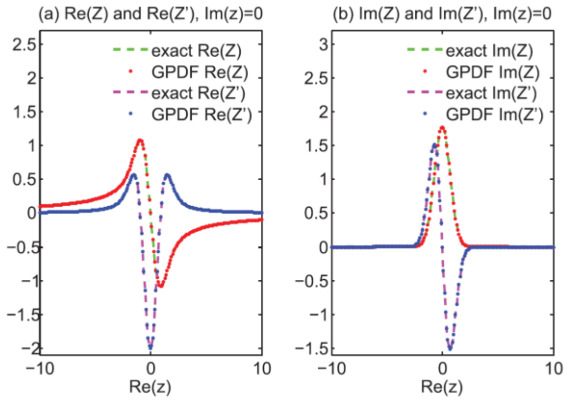



FIG. 2. Comparison of our scheme for GPDF with exact $Z(\zeta)$ function via complex error function in standard numerical library on real axis.

standard numerical library on real axis. We find the errors to be around $10^{-12}$ (not shown in the figure), e.g., $Z(1) = -1.076159013825734 + 0.652049332173291i$ in our scheme and $Z(1) = -1.076159013825537 + 0.652049332173292i$ via standard library.

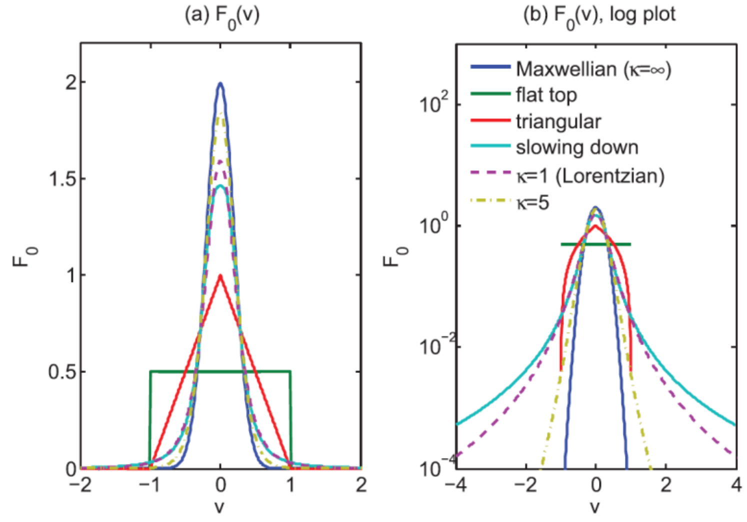



FIG. 1. Typical distribution functions.

Generalized plasma dispersion function (GPDF). Support almost arbitrary distributions, N=32 for most cases.

Feature: Fast, accurate, exponential convergent (relevant to FFT, Weideman94), rational form (useful, later)

Also, Robison90 (Hermite, Legendre, or Chebyshev), Bilato14

Phys. Plasmas 20, 092125 (2013)

https://github.com/hsxie/gpdf

# 2. Matrix method to obtain all solutions of fluid waves

Computer Physics Communications 185 (2014) 670–675.

## 2.1 From k to omega (Xie2014CPC)

**2. Theory and method**

We consider a multi-fluid plasma in an external magnetic field $\mathbf{B}_0 = (0, 0, B_0)$. The flow velocity of the fluid component $j$ is $\mathbf{v}_{j0} = (v_{j0x}, v_{j0y}, v_{j0z})$. The species densities are allowed to be locally inhomogeneous, with $\nabla n_{j0}/n_{j0} = (\epsilon_{njx}, \epsilon_{njy}, 0) =$ constant. For simplicity, temperature gradient effects are ignored, and the wave vector is assumed to be $\mathbf{k} = (k_x, 0, k_z) = (k\sin\theta, 0, k\cos\theta)$. We do not need to obtain the 3 by 3 dispersion relation matrix for $\mathbf{E} = (E_x, E_y, E_z)$, as done in many existing analytical or numerical treatments, such as in Stix [4], Ronnmark [1,2] and Bret [3]. Instead, we use the original full dispersion relation matrix and treat it as a matrix eigenvalue problem, instead of directly solving for its determinant.

*2.1. Governing equations*

We start with the fluid equations

$$\partial_t n_j = -\nabla\cdot(n_j\mathbf{v}_j), \tag{1a}$$

$$\partial_t \mathbf{u}_j = -\mathbf{v}_j\cdot\nabla\mathbf{u}_j + \frac{q_j}{m_j}(\mathbf{E}+\mathbf{v}_j\times\mathbf{B}) - \frac{\nabla\mathbf{P}_j}{\rho_j} - \sum_i(\mathbf{u}_i-\mathbf{u}_j)\nu_{ij}, \tag{1b}$$

$$\partial_t\mathbf{E} = c^2\nabla\times\mathbf{B} - \mathbf{J}/\epsilon_0, \tag{1c}$$

$$\partial_t\mathbf{B} = -\nabla\times\mathbf{E}, \tag{1d}$$

where $\mathbf{u}_j = \gamma_j\mathbf{v}_j$, and

$$\mathbf{J} = \sum_j q_j n_j\mathbf{v}_j, \tag{2a}$$

$$d_t(P_{\parallel j}\rho_j^{-\gamma_{\parallel j}}) = 0, \tag{2b}$$

$$d_t(P_{\perp j}\rho_j^{-\gamma_{\perp j}}) = 0, \tag{2c}$$

where $\rho_j \equiv m_j n_j$, $c^2 = 1/\mu_0\epsilon_0$, $\gamma_j = (1-v_j^2/c^2)^{-1/2}$, and $\gamma_{\parallel j}$ and $\gamma_{\perp j}$ are the parallel and perpendicular adiabatic coefficients, respectively. Furthermore, $P_{\parallel,\perp} = nT_{\parallel,\perp}$, $\mathbf{P} = P_\parallel\hat{\mathbf{b}}\hat{\mathbf{b}} + P_\perp(\mathbf{I}-\hat{\mathbf{b}}\hat{\mathbf{b}})$ and $\hat{\mathbf{b}} = \mathbf{B}/B$. Note that our anisotropy model differs from that of CGL [6], but can be reduced to that of Bret and Deutsch [5] by setting $\gamma_{\parallel j} = \gamma_{\perp j} = \gamma_{Tj}$. By further setting $T_{\perp j} = T_{\parallel j}$, we can recover the isotropic pressure case.

After linearizing, (2) becomes

$$\mathbf{J} = \sum_j q_j(n_{j0}\mathbf{v}_{j1} + n_{j1}\mathbf{v}_{j0}), \tag{3a}$$

*2.2. Full-matrix treatment*

The linearized version of (1) with $f = f_0 + f_1 e^{i\mathbf{k}\cdot\mathbf{r}-i\omega t}$, $f_1 \ll f_0$ is equivalent to a matrix eigenvalue problem

$$\lambda\mathbf{AX} = \mathbf{MX}, \tag{5}$$

with $\lambda = -i\omega$ the eigenvalue and $\mathbf{X}$ the corresponding eigenvector, which also gives the polarization of each normal/eigenmode solution. Similar treatments can be found in [7] for the MHD equations and [8] for the ten-moment equations.

Accordingly, we have $\mathbf{X} = (n_{j1}, v_{j1x}, v_{j1y}, v_{j1z}, E_{1x}, E_{1y}, E_{1z}, B_{1x}, B_{1y}, B_{1z})^T$, $\mathbf{u}_{j1} = \gamma_{j0}[\mathbf{v}_{j1} + \gamma_{j0}^2(\mathbf{v}_{j0}\cdot\mathbf{v}_{j1})\mathbf{v}_{j0}/c^2] = \{a_{jpq}\}\cdot\mathbf{v}_{j1}$ $(p, q = x, y, z)$, $\gamma_{j0} = (1-v_{j0}^2/c^2)^{-1/2}$, and $\mathbf{A}$ is given by

$$\begin{bmatrix} \{1 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0\\ 0 & a_{jxx} & a_{jxy} & a_{jxz} & 0 & 0 & 0 & 0 & 0 & 0\\ 0 & a_{jyx} & a_{jyy} & a_{jyz} & 0 & 0 & 0 & 0 & 0 & 0\\ 0 & a_{jzx} & a_{jzy} & a_{jzz}\} & 0 & 0 & 0 & 0 & 0 & 0\\ 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 & 0\\ 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0\\ 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0\\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0\\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0\\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 \end{bmatrix}. \tag{6}$$

For simplicity, the relativistic effects in the friction terms $\nu_{ij}$ are ignored. The matrix $\mathbf{M}$ is then given by ($\nu_{ij}$ terms when $i \neq j$ are not given explicitly here) Eq. (7) (see Box I).

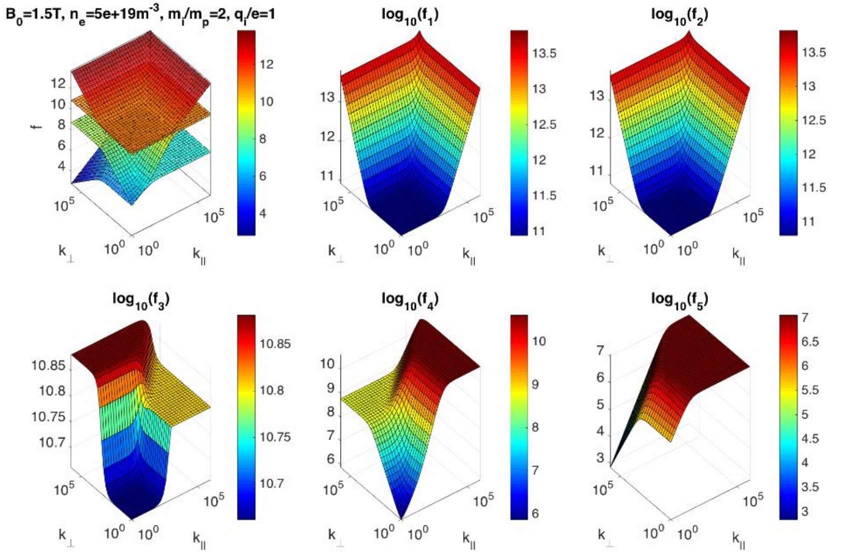
Furthermore,

$$\{a_{jpq}\} \equiv \begin{bmatrix} a_{jxx} & a_{jxy} & a_{jxz}\\ a_{jyx} & a_{jyy} & a_{jyz}\\ a_{jzx} & a_{jzy} & a_{jzz} \end{bmatrix} = \begin{bmatrix} \gamma_{j0}+\gamma_{j0}^3 v_{j0x}^2/c^2 & \gamma_{j0}^3 v_{j0x}v_{j0y}/c^2 & \gamma_{j0}^3 v_{j0x}v_{j0z}/c^2\\ \gamma_{j0}^3 v_{j0x}v_{j0y}/c^2 & \gamma_{j0}+\gamma_{j0}^3 v_{j0y}^2/c^2 & \gamma_{j0}^3 v_{j0y}v_{j0z}/c^2\\ \gamma_{j0}^3 v_{j0x}v_{j0z}/c^2 & \gamma_{j0}^3 v_{j0z}v_{j0y}/c^2 & \gamma_{j0}+\gamma_{j0}^3 v_{j0z}^2/c^2 \end{bmatrix}. \tag{8}$$

For a plasma containing $s$ species of particles, the dimensions of $\mathbf{A}$ and $\mathbf{M}$ are $(4s+6)\times(4s+6)$. We can get all the linear harmonic wave solutions of the system without any convergence difficulty using a standard matrix eigenvalue solver, e.g., LAPACK or the function *eig()* in *MATLAB*. Here, a *MATLAB* code PDRF for solving the above eigenvalue problem is provided. By setting $\gamma_j$ to 1, i.e., $\mathbf{A} = \mathbf{I}$ and $\{a_{jpq}\} = \mathbf{I}$, PDRF reduces to the non-relativistic case.

$$\begin{bmatrix} \{-i\mathbf{k}\cdot\mathbf{v}_{j0} & -ik_x n_{j0}-\epsilon_{njx}n_{j0} & -\epsilon_{njy}n_{j0} & -ik_z n_{j0} & 0 & 0 & 0 & 0 & 0 & 0\\ \frac{-ik_x c_{\perp j}^2}{\rho_{j0}} & b_{jxx} & b_{jxy}+\omega_{cj} & b_{jxz} & \frac{q_j}{m_j} & 0 & 0 & -\frac{ik_z\Delta_j}{m_j n_{j0}} & -\frac{q_j v_{j0z}}{m_j} & \frac{q_j v_{j0y}}{m_j}\\ 0 & b_{jyx}-\omega_{cj} & b_{jyy} & b_{jyz} & 0 & \frac{q_j}{m_j} & 0 & \frac{q_j v_{j0z}}{m_j} & -\frac{ik_z\Delta_j}{m_j n_{j0}} & -\frac{q_j v_{j0x}}{m_j}\\ \frac{-ik_z c_{\parallel j}^2}{\rho_{j0}} & b_{jzx} & b_{jzy} & b_{jzz}\} & 0 & 0 & \frac{q_j}{m_j} & -\frac{q_j v_{j0y}}{m_j}-\frac{ik_x\Delta_j}{m_j n_{j0}} & \frac{q_j v_{j0x}}{m_j} & 0\\ -\frac{q_j v_{j0x}}{\epsilon_0} & -\frac{q_j n_{j0}}{\epsilon_0} & 0 & 0 & 0 & 0 & 0 & 0 & -ik_z c^2 & 0\\ -\frac{q_j v_{j0y}}{\epsilon_0} & 0 & -\frac{q_j n_{j0}}{\epsilon_0} & 0 & 0 & 0 & 0 & ik_z c^2 & 0 & -ik_x c^2\\ -\frac{q_j v_{j0z}}{\epsilon_0} & 0 & 0 & -\frac{q_j n_{j0}}{\epsilon_0} & 0 & 0 & 0 & 0 & ik_x c^2 & 0\\ 0 & 0 & 0 & 0 & 0 & ik_z & 0 & 0 & 0 & 0\\ 0 & 0 & 0 & 0 & -ik_z & 0 & ik_x & 0 & 0 & 0\\ 0 & 0 & 0 & 0 & 0 & -ik_x & 0 & 0 & 0 & 0 \end{bmatrix},$$

e $\omega_{cj} = q_jB_0/m_j$, $q_e = -e$, $\omega_{pj}^2 = n_{j0}q_j^2/\epsilon_0 m_j$, and $\{b_{jpq}\} = \nu_{jj} - i(\mathbf{k}\cdot\mathbf{v}_{j0})\cdot\{a_{jpq}\}$.




Feature: Fast, accurate, obtain all solutions and polarizations directly, valid for arbitrary fluid models

https://github.com/hsxie/pdrf

# 2. Matrix method to obtain all solutions of fluid waves

## 2.2 From omega to k (Xie2024FPP)

*3.2. Warm multi-fluid plasma dispersion equations*

The multi-fluid equations are [5,10]

$$\partial_t n_s = -\nabla\cdot(n_s \boldsymbol{v}_s), \tag{5a}$$

$$\partial_t \boldsymbol{v}_s = -\boldsymbol{v}_s\cdot\nabla\boldsymbol{v}_s + \frac{q_s}{m_s}(\boldsymbol{E}+\boldsymbol{v}_s\times\boldsymbol{B}) - \frac{\nabla P_s}{\rho_s}, \tag{5b}$$

$$\partial_t \boldsymbol{E} = c^2\nabla\times\boldsymbol{B} - \boldsymbol{J}/\epsilon_0, \tag{5c}$$

$$\partial_t \boldsymbol{B} = -\nabla\times\boldsymbol{E}, \tag{5d}$$

where the current and pressure equations are

$$\boldsymbol{J} = \sum_s q_s n_s \boldsymbol{v}_s, \tag{6a}$$

$$d_t(P_s\rho_s^{-\gamma_s}) = 0, \tag{6b}$$

with the mass density $\rho_s \equiv m_s n_s$. According to Ref. [11], to be close to the results of kinetic model, the adiabatic coefficient is set $\gamma_s = 2$ instead of $\gamma_s = 5/3$.

Linearizing the equations using $f = f_0 + \delta f$ and $\delta f = \delta f e^{i(\boldsymbol{k}\cdot\boldsymbol{r}-\omega t)}$, we obtain

$$\delta\boldsymbol{J} = \sum_s q_s n_s \boldsymbol{v}_s, \tag{7a}$$

$$\delta P_s = P_{s0}\gamma_s\delta\rho_s = c_s^2 m_s\delta n_s, \tag{7b}$$

where $c_s^2 \equiv \gamma_s P_{s0}/\rho_{s0} = \gamma_s k_B T_{s0}/m_s$ and $P_{s0} = n_{s0}k_B T_{s0}$, and thus

$$\nabla\delta P_s = (ik_x\delta P_s, 0, ik_z\delta P_s). \tag{8}$$

However, in this work, we are interested in the solutions of $\boldsymbol{k}$, especially $k_x$, for a given $\omega$ and other parameters. Eqs. (9) can be rewritten as

$$k_x c_s^2\delta n_s = \omega n_{s0}\delta v_{sx} - \frac{n_{s0}q_s}{\omega m_s}\Big[ik_z c^2\delta B_y + \frac{\sum_{s'} q_{s'}n_{s'0}\delta v_{s'x}}{\epsilon_0} - \omega_{cs}(\delta E_y - \delta v_{sx}B_0)\Big], \tag{10a}$$

$$k_x n_{s0}\delta v_{sx} = \omega\delta n_s - \left(i\frac{q_s}{m_s}\frac{k_z n_{s0}}{\omega}\delta E_z + \frac{k_z^2 c_s^2}{\omega}\delta n_s\right), \tag{10b}$$

$$k_x\delta E_y = \omega\delta B_z, \tag{10c}$$

$$k_x\delta E_z = \left(\frac{k_z^2c^2}{\omega} - \omega\right)\delta B_y - i\frac{k_z}{\omega}\frac{\sum_{s'} q_{s'}n_{s'0}\delta v_{s'x}}{\epsilon_0}, \tag{10d}$$

$$k_x c^2\delta B_y = -\omega\delta E_z + \frac{1}{\omega}\sum_{s'}\left(\omega_{ps'}^2\delta E_z - i\frac{k_z c_{s'}^2 q_{s'}}{\epsilon_0}\delta n_{s'}\right), \tag{10e}$$

$$k_x c^2\delta B_z = \left(\omega - \frac{k_z^2c^2}{\omega}\right)\delta E_y - \sum_{s'}\frac{\omega_{ps'}^2}{\omega}(\delta E_y - \delta v_{s'x}B_0), \tag{10f}$$

where $2S+4$ equations for $k_x$ are obtained and thus have $2S+4$ solutions of $k_x$ for a given other parameters. Note also that in the first equation, $(n_{s0}q_s/m_s)(\sum_{s'} q_{s'}n_{s'0}\delta v_{s'x}/\epsilon_0) \neq \sum_{s'} q_{s'}^2 n_{s'0}^2\delta v_{s'x}/(m_{s'}\epsilon_0)$. In the cold plasma limit, $c_s^2 = 0$, the above equations can be reduced to 4 equations and thus have 4 branches of $k_x$ in the cold plasma, which agrees with the standard case in Section 3.1.

Eqs. (10) can readily yield a matrix eigenvalue equation

$$k_x\cdot\boldsymbol{X} = \boldsymbol{M}\cdot\boldsymbol{X}, \tag{11}$$

with $\boldsymbol{X} = [\delta n_s, \delta v_{sx}, \delta E_y, \delta E_z, \delta B_y, \delta B_z]^T$. The numerical solutions of all the eigenvalues $k_x$ and corresponding eigenvectors $\boldsymbol{X}$ can be easily obtained by a standard eigen matrix solver. The eigenvector $\boldsymbol{X}$ represents the polarizations of the wave. Use the above model, all the wave solutions of multi-fluid plasmas with temperature effects are obtained.

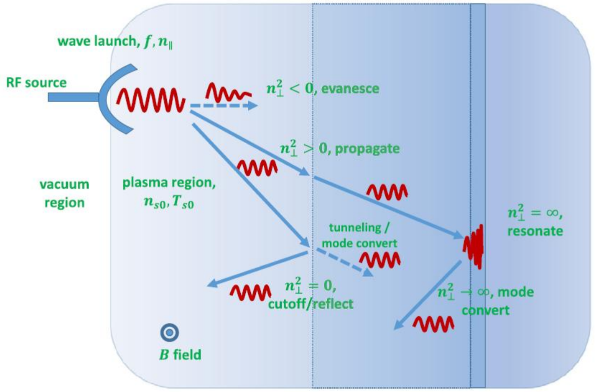



**Fig. 1.** A cartoon to show the plasma wave propagation, which represents the wave accessibility or propagation conditions.

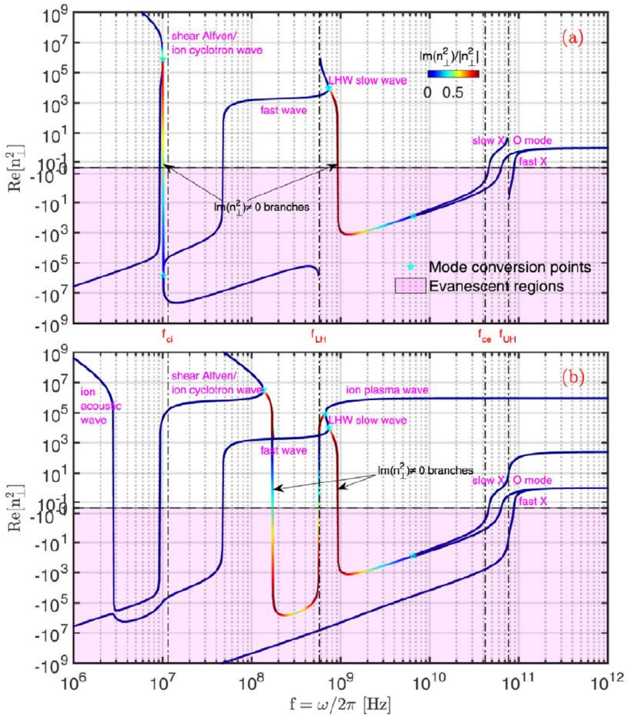



**Fig. 3.** The dispersion relations of the warm multi-fluid and cold plasma with using the same parameters as in Fig. 2, for $n_\perp^2$ vs $f$.

Fundamental Plasma Physics 10 (2024) 100050.

Feature: Helpful to fast obtain all fluid solutions of plasma wave propagation conditions/accessibility

https://github.com/hsxie/bon

# 3. Obtain all solutions of kinetic dispersion relation

## 3.1 Electrostatic 1D case (key ideas)

### 2.1 Electrostatic 1D

First, we solve the simplest multi-component electrostatic 1D (ES1D) problem with drift Maxwellian distribution $f_{s0} = (\frac{m_s}{2\pi k_B T_s})^{1/2} \exp[-\frac{(v-v_{s0})^2}{2k_B T_s}]$. The dispersion relation is

$$D = 1 + \sum_{s=1}^{S} \frac{1}{(k\lambda_{Ds})^2}[1 + \zeta_s Z(\zeta_s)] = 0, \qquad (1)$$

where $\lambda_{Ds}^2 = \frac{\epsilon_0 k_B T_s}{n_s q_s^2}$, $v_{ts} = \sqrt{\frac{2k_B T_s}{m_s}}$ and $\zeta_s = \frac{\omega - kv_{s0}}{kv_{ts}}$. Unmentioned notations are standard. The plasma dispersion function $Z(\zeta) = \frac{1}{\sqrt{\pi}}\int_{-\infty}^{\infty}\frac{e^{-z^2}}{z-\zeta}dz$ can be approximated using $J$-pole expansion

$$Z(\zeta) \simeq Z_J(\zeta) = \sum_{j=1}^{J} \frac{b_j}{\zeta - c_j}, \qquad (2)$$

where $J = 8$ is used by Ronnmark [2,3] and $J = 2, 3, 4$ are provided by Martin et al. [10], producing accurate results for most domains (except $y < \sqrt{\pi}x^2 e^{-x^2}$ when $x \gg 1$, with $\zeta = x + iy$), especially in the upper plane. However, the method does not perform well for heavily damped modes, which are of little interest anyway. For completeness, the coefficients $c_j$ and $b_j$ for $J = 4$, $J = 8$ and $J = 12$ (see Appendix A) are provided in Table 1. Note the useful relations [2] $\sum_j b_j = -1$, $\sum_j b_j c_j = 0$ and $\sum_j b_j c_j^2 = -1/2$.

Combining Eqs. (1) and (2), yields

$$1 + \sum_s \sum_j \frac{b_{sj}}{(\omega - c_{sj})} = 0, \qquad (3)$$

with $b_{sj} = \frac{b_j c_j v_{ts}}{k\lambda_{Ds}^2}$ and $c_{sj} = k(v_{s0} + v_{ts}c_j)$. The solutions for the frequency $\omega = \omega_r + i\omega_i = \omega_r + i\gamma$ are usually complex numbers. An equivalent linear system can be obtained as follows:

$$\omega n_{sj} = c_{sj} n_{sj} + b_{sj} E, \qquad (4a)$$

$$E = -\sum_{sj} n_{sj}, \qquad (4b)$$

which is an eigenvalue problem of a $SJ \times SJ$ dimensional eigen matrix $\boldsymbol{M}$, i.e., $\omega \boldsymbol{X} = \boldsymbol{M}\boldsymbol{X}$, with $SJ = S \times J$ and $\boldsymbol{X} = \{n_{sj}\}$. The symbols $n_{sj}$ and $E$ used here do not have direct physical meanings but are analogy to the perturbed number density and electric field in fluid derivation of Langmuir wave. The singularity in the denominator of (3), which is encountered in conventional methods, can be canceled by using the transformation (4). Hence, the matrix method can easily support multi-component systems.

Feature: Fast, accurate, obtain all (important) solutions (except less important strong damped modes) of a kinetic system!

Plasma Science and Technology, 18, 2, 97 (2016).

Key steps:

- Rational expansion (J-pole, high accuracy)
- Linear matrix transformation

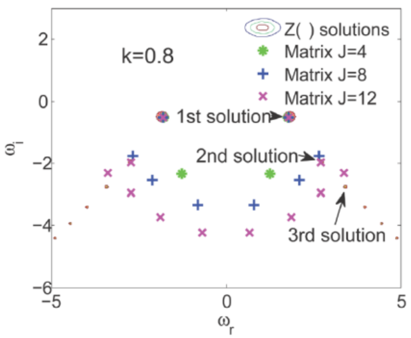



**Fig.1** Comparison of all the solutions obtained from the matrix method and the $Z(\zeta)$ function. To avoid missing solutions, the $Z(\zeta)$ function solutions are shown by contour plot of Eq. (1) $|D(\omega_r, \omega_i)| = 0$ in the complex plane, which can show the distribution of all solutions in the complex plane directly. The arrows in the figures point out the positions of the first three solutions. We can see that $J = 4, 8, 12$ all can reproduce the first solution, but only $J = 12$ can reproduce the second solution accurately

| $k\lambda_{De}$ | $\omega_r^M(J=4)$ | $\omega_i^M(J=4)$ | $\omega_r^M(J=8)$ | $\omega_i^M(J=8)$ | $\omega_r^M(J=12)$ | $\omega_i^M(J=12)$ | $\omega_r^Z$ | $\omega_i^Z$ |
|---|---|---|---|---|---|---|---|---|
| 0.1 | 0.9956 | 9.5E-3 | 1.0152 | 1.7E−5 | 1.0152 | 9.5E−8 | 1.0152 | −4.8E−15 |
| 0.5 | 1.4235 | −0.1699 | 1.4156 | −0.1534 | 1.4157 | −0.1534 | 1.4157 | −0.1534 |
| 1.0 | 2.0170 | −0.8439 | 2.0459 | −0.8514 | 2.0458 | −0.8513 | 2.0458 | −0.8513 |
| 2.0 | 3.2948 | −2.6741 | 3.1893 | −2.8272 | 3.1891 | −2.8272 | 3.1891 | −2.8272 |

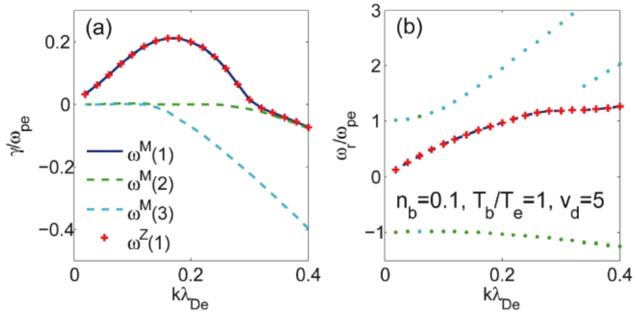



**Fig.2** Comparison of the first three ($\omega^M$) largest imaginary part solutions obtained from the matrix method ($J = 8$) and one solution ($\omega^Z$) obtained from $Z(\zeta)$ function for the bump-on-tail parameters. The matrix method reproduces one of the $Z(\zeta)$ function solutions accurately and can also give other solutions directly

https://github.com/hsxie/pdrk

# 3. Obtain all solutions of kinetic dispersion relation

## 3.2 Drift bi-Maxwellian case (Xie2016PST)

The background magnetic field is assumed to be $\boldsymbol{B}_0 = (0, 0, B_0)$, and the wave vector $\boldsymbol{k} = (k_x, 0, k_z) = (k\sin\theta, 0, k\cos\theta)$, which gives $k_\perp = k_x$ and $k_\parallel = k_z$.

The dispersion relation is [Stix1992, Miyamoto2004]

$$\begin{vmatrix} K_{xx} - \frac{c^2k^2}{\omega^2}\cos^2\theta & K_{xy} & K_{xz} + \frac{c^2k^2}{\omega^2}\sin\theta\cos\theta \\ K_{yx} & K_{yy} - \frac{c^2k^2}{\omega^2} & K_{yz} \\ K_{zx} + \frac{c^2k^2}{\omega^2}\sin\theta\cos\theta & K_{zy} & K_{zz} - \frac{c^2k^2}{\omega^2}\sin^2\theta \end{vmatrix} = 0, \qquad (1)$$

with $\boldsymbol{K} = \boldsymbol{I} + \sum_s \frac{\omega_{ps}^2}{\omega^2}\left[\sum_n \{\zeta_0 Z(\zeta_n) - (1 - \frac{1}{\lambda_T})[1 + \zeta_n Z(\zeta_n)]\}\boldsymbol{X}_n + 2\eta_0^2\lambda_T\boldsymbol{L}\right]$, where in the original paper [Xie2016]

$$\boldsymbol{X}_n = \begin{pmatrix} n^2\Gamma_n/b & in\Gamma_n' & (2\lambda_T)^{1/2}\eta_n\frac{n}{\alpha}\Gamma_n \\ -in\Gamma_n' & n^2\Gamma_n/b - 2b\Gamma_n' & -i(2\lambda_T)^{1/2}\eta_n\alpha\Gamma_n' \\ (2\lambda_T)^{1/2}\eta_n\frac{n}{\alpha}\Gamma_n & i(2\lambda_T)^{1/2}\eta_n\alpha\Gamma_n' & 2\lambda_T\eta_n^2\Gamma_n \end{pmatrix}$$

### 1.3 The linear transformation

To seek an equivalent linear system, the Maxwells equations

$$\partial_t\boldsymbol{E} = c^2\nabla\times\boldsymbol{B} - \boldsymbol{J}/\epsilon_0, \qquad (4a)$$

$$\partial_t\boldsymbol{B} = -\nabla\times\boldsymbol{E}, \qquad (4b)$$

do not need to be changed. We only need to seek a new linear system for $\boldsymbol{J} = \overleftrightarrow{\sigma}\cdot\boldsymbol{E}$. It is easy to find that after $J$-pole expansion, the relations between $\boldsymbol{J}$ and $\boldsymbol{E}$ has the following form

$$\begin{pmatrix} J_x \\ J_y \\ J_z \end{pmatrix} = -i\epsilon_0 \begin{pmatrix} \frac{b_{11}}{\omega} + \sum_{snj}\frac{b_{snj11}}{\omega - c_{snj}} & \frac{b_{12}}{\omega} + \sum_{snj}\frac{b_{snj12}}{\omega - c_{snj}} & \frac{b_{13}}{\omega} + \sum_{snj}\frac{b_{snj13}}{\omega - c_{snj}} \\ \frac{b_{21}}{\omega} + \sum_{snj}\frac{b_{snj21}}{\omega - c_{snj}} & \frac{b_{22}}{\omega} + \sum_{snj}\frac{b_{snj22}}{\omega - c_{snj}} & \frac{b_{23}}{\omega} + \sum_{snj}\frac{b_{snj23}}{\omega - c_{snj}} \\ \frac{b_{31}}{\omega} + \sum_{snj}\frac{b_{snj31}}{\omega - c_{snj}} & \frac{b_{32}}{\omega} + \sum_{snj}\frac{b_{snj32}}{\omega - c_{snj}} & \frac{b_{33}}{\omega} + \sum_{snj}\frac{b_{snj33}}{\omega - c_{snj}} \end{pmatrix} \begin{pmatrix} E_x \\ E_y \\ E_z \end{pmatrix}. \qquad (6)$$

Combining Eqs. (4) and (6), the equivalent linear system for (1) can be obtained as[3]

$$\begin{cases} \omega v_{snjx} = c_{snj}v_{snjx} + b_{snj11}E_x + b_{snj12}E_y + b_{snj13}E_z, \\ \omega j_x = b_{11}E_x + b_{12}E_y + b_{13}E_z, \\ iJ_x/\epsilon_0 = j_x + \sum_{snj} v_{snjx}, \\ \omega v_{snjy} = c_{snj}v_{snjy} + b_{snj21}E_x + b_{snj22}E_y + b_{snj23}E_z, \\ \omega j_y = b_{21}E_x + b_{22}E_y + b_{23}E_z, \\ iJ_y/\epsilon_0 = j_y + \sum_{snj} v_{snjy}, \\ \omega v_{snjz} = c_{snj}v_{snjz} + b_{snj31}E_x + b_{snj32}E_y + b_{snj33}E_z, \\ \omega j_z = b_{31}E_x + b_{32}E_y + b_{33}E_z, \\ iJ_z/\epsilon_0 = j_z + \sum_{snj} v_{snjz}, \\ \omega E_x = +c^2k_zB_y - iJ_x/\epsilon_0, \\ \omega E_y = -c^2k_zB_x + c^2k_xB_z - iJ_y/\epsilon_0, \\ \omega E_z = -c^2k_xB_y - iJ_z/\epsilon_0, \\ \omega B_x = -k_zE_y, \\ \omega B_y = +k_zE_x - k_xE_z, \\ \omega B_z = +k_xE_y, \end{cases} \qquad (7)$$

which yields a sparse matrix eigenvalue problem. Again, the symbols $v_{snjx}$, $j_{x,y,z}$ and $J_{x,y,z}$ used here do not have direct physical meanings but are analogy to the perturbed velocity and current density in the fluid derivations of plasma waves. The elements of the eigenvector $(E_x, E_y, E_z, B_x, B_y, B_z)$ still represent the original electric and magnetic fields. Thus, the polarization of the solutions can also be obtained in a straightforward manner. The dimension of the matrix is $NN = 3\times(SNJ + 1) + 6 = 3\times[S\times(2\times N + 1)\times J + 1] + 6$. The coefficients in original paper [Xie2014] are

Plasma Science and Technology, 18, 2, 97 (2016).

Similar to ES1D case, a much complicated but tractable derivation.

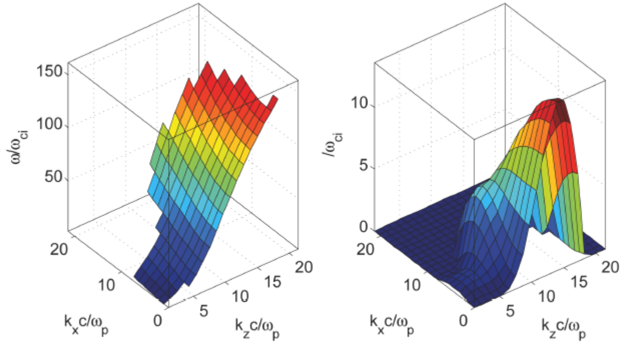



**Fig.10** Electromagnetic whistler beam instability. The real frequency $\omega$ is only shown for unstable ($\gamma > 0$) solutions. The parallel propagation ($k = k_\parallel$) results are similar to Fig.8.8 of Ref. [5]. $N = 3$ is used for this calculation

Ronnmark82, WHAMP, using 8-pole, but iterative root finding, require initial guess

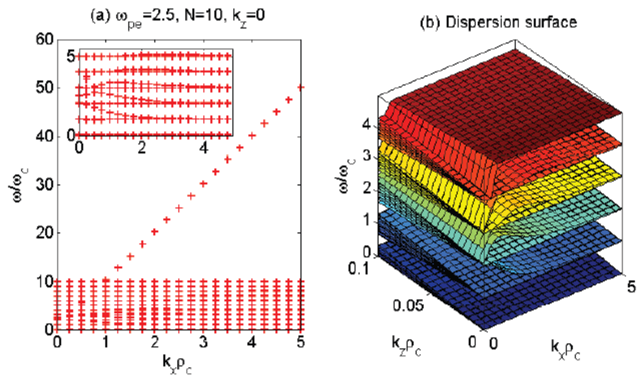



**Fig.11** Dispersion surface (b) from PDRK-EM3D, using the EBW parameters in Fig. 3 and $c^2 = 10^2$. The $\omega$ vs. $k_\perp$ (a) result is close to the ES3D result in Fig. 3, which confirms that EBW is (quasi-) electrostatic

Feature: Fast, accurate, obtain all solutions (except less important strong damped modes) of EM3D kinetic system. Without singularity, no need initial guess for root finding. The first time we have such a solver!

https://github.com/hsxie/pdrk

# 3. Obtain all solutions of kinetic dispersion relation

## 3.3 With ring beam, perpendicular drifts, collision, etc (Xie2019CPC)

We consider only the collisionless case or with a Krook collision, the kinetic equation for each species is the Vlasov equation with Krook collision at the right hand side

$$\frac{\partial F_s}{\partial t} + \boldsymbol{v}\cdot\frac{\partial F_s}{\partial \boldsymbol{r}} + \left[\boldsymbol{a}_s + \frac{q_s}{m_s}(\boldsymbol{E}+\boldsymbol{v}\times\boldsymbol{B})\right]\cdot\frac{\partial F_s}{\partial \boldsymbol{v}} = -\nu_s(F_s - F_{s0}), \tag{1}$$

where the distribution function $F_s(\boldsymbol{r},\boldsymbol{v},t) = n_{s0}f_s(\boldsymbol{r},\boldsymbol{v},t)$, and $q_s$, $m_s$ and $n_{s0}$ are the charge, mass and the number density of species $s$, respectively. When the collision frequency $\nu_s = 0$, the equation reduces to a collisionless case. The Maxwell equations for fields can be either

$$\partial_t\boldsymbol{E} = c^2\nabla\times\boldsymbol{B} - \boldsymbol{J}/\epsilon_0, \tag{2}$$

$$\partial_t\boldsymbol{B} = -\nabla\times\boldsymbol{E}, \tag{3}$$

for electromagnetic case, or

$$\nabla\cdot\boldsymbol{E} = \rho/\epsilon_0, \tag{4}$$

$$\boldsymbol{E} = -\nabla\Phi, \tag{5}$$

**3. The dispersion relation for extended Maxwellian distribution**

The extended Maxwellian distribution here means bi-Maxwellian distribution with loss cone, parallel and perpendicular drifts and ring beam. We model it using the following equilibrium distribution function.

*3.1. Equilibrium distribution function*

We assume equilibrium distribution function $F_{s0}(v'_\parallel, v'_\perp) = n_{s0}f_{s0}(v'_\parallel, v'_\perp)$, with $v'_\parallel = v_z$, $v'_\perp = \sqrt{(v_x - v_{dsx})^2 + (v_y - v_{dsy})^2}$, and

$$f_{s0}(v'_\parallel, v'_\perp) = f_{s0z}(v'_\parallel)f_{s0\perp}(v'_\perp) \tag{64}$$

$$= \frac{1}{\pi^{3/2}v_{zts}v_{\perp ts}^2}\exp\left[-\frac{(v'_\parallel - v_{dsz})^2}{v_{zts}^2}\right]\left\{\frac{r_{sa}}{A_{sa}}\exp\left[-\frac{(v'_\perp - v_{dsr})^2}{v_{\perp ts}^2}\right] + \frac{r_{sb}}{\alpha_s A_{sb}}\exp\left[-\frac{(v'_\perp - v_{dsr})^2}{\alpha_s v_{\perp ts}^2}\right]\right\},$$

where $r_{sa} = \left(\frac{1-\alpha_s\Delta_s}{1-\alpha_s}\right)$ and $r_{sb} = \left(\frac{-\alpha_s+\alpha_s\Delta_s}{1-\alpha_s}\right)$, and

$$A_{s\sigma} = \exp\left(-\frac{v_{dsr}^2}{v_{\perp ts\sigma}^2}\right) + \frac{\sqrt{\pi}\,v_{dsr}}{v_{\perp ts\sigma}}\mathrm{erfc}\left(-\frac{v_{dsr}}{v_{\perp ts\sigma}}\right), \quad \text{for } \sigma = a, b, \tag{65}$$

with $v_{\perp tsa} = v_{\perp ts}$ and $v_{\perp tsb} = \sqrt{\alpha_s}v_{\perp ts}$.

- Beyond the Stix gyrotropic distributions to including perpendicular drifts & collision
- Complicated but tractable derivations



*4.2.1. The unmagnetized terms*

Considering the definition $\boldsymbol{\sigma}_s^u = -i\epsilon_0\omega\boldsymbol{Q}_s^u = -i\epsilon_0\sum_{\sigma=a,b} r_{s\sigma}\frac{\omega_{ps}^2}{\omega}\boldsymbol{P}_{s\sigma}^u$, after $J$-pole expansion, we have

- $P^u_{s\sigma 11} = -1 + \frac{2}{kv_{ts\sigma}}\Big\{-i\nu_s\big(\frac{k_x^2v_{\perp ts\sigma}^2}{k^2v_{ts\sigma}^2}Z_2 + \frac{k_xv_{dsx}}{kv_{ts\sigma}}Z_1 + \frac{k_z^2v_{zts}^2}{2k^2v_{ts\sigma}^2}Z_0\big) + \frac{k^2k_x^2v_{\perp ts\sigma}^4}{k^3v_{ts\sigma}^3}Z_3 + 2\frac{k^2k_xv_{\perp ts\sigma}^2}{k^2v_{ts\sigma}^2}v_{dsx}Z_2 + [\frac12 v_{\perp ts\sigma}^2\frac{k^2k_z^2v_{zts}^2}{k^3v_{ts\sigma}^3} + \frac{k^2}{kv_{ts\sigma}}v_{dsx}^2 + (\lambda_{Ts\sigma}-1)\frac{k_x^2k_z^2v_{\perp ts\sigma}^4}{k^3v_{ts\sigma}^3}]Z_1 + \frac{k_xk_z^2}{k^2v_{ts\sigma}^2}(\lambda_{Ts\sigma}-1)v_{\perp ts\sigma}^2v_{dsx}Z_0\Big\} \simeq -1 + 2\sum_{j=1}^J\frac{b_j}{\omega-c_{sj\sigma}}\Big\{-i\nu_s\big(\frac{k_x^2v_{\perp ts\sigma}^2}{k^2v_{ts\sigma}^2}c_j^2 + \frac{k_xv_{dsx}}{kv_{ts\sigma}}c_j + \frac{k_z^2v_{zts}^2}{2k^2v_{ts\sigma}^2}\big) + \frac{k^2k_x^2v_{\perp ts\sigma}^4}{k^3v_{ts\sigma}^3}c_j^3 + 2\frac{k^2k_xv_{\perp ts\sigma}^2}{k^2v_{ts\sigma}^2}v_{dsx}c_j^2 + [\frac12 v_{\perp ts\sigma}^2\frac{k^2k_z^2v_{zts}^2}{k^3v_{ts\sigma}^3} + \frac{k^2}{kv_{ts\sigma}}v_{dsx}^2 + (\lambda_{Ts\sigma}-1)\frac{k_x^2k_z^2v_{\perp ts\sigma}^4}{k^3v_{ts\sigma}^3}]c_j + \frac{k_xk_z^2}{k^2v_{ts\sigma}^2}(\lambda_{Ts\sigma}-1)v_{\perp ts\sigma}^2v_{dsx}\Big\} = -1 + \sum_{j=1}^J\frac{p_{11sj\sigma}}{\omega-c_{sj\sigma}}$,
- $P^u_{s\sigma 12} = \frac{2v_{dsy}}{kv_{ts\sigma}}\Big\{\frac{k^2k_xv_{\perp ts\sigma}^2}{k^2v_{ts\sigma}^2}Z_2 + \frac{k^2}{kv_{ts\sigma}}v_{dsx}Z_1 + \frac12\frac{k_xk_z^2}{k^2v_{ts\sigma}^2}(\lambda_{Ts\sigma}-1)v_{\perp ts\sigma}^2Z_0\Big\} \simeq 2v_{dsy}\sum_{j=1}^J\frac{b_j}{\omega-c_{sj\sigma}}\Big\{\frac{k^2k_xv_{\perp ts\sigma}^2}{k^2v_{ts\sigma}^2}c_j^2 + \frac{k^2v_{dsx}}{kv_{ts\sigma}}c_j + \frac12\frac{k_xk_z^2}{k^2v_{ts\sigma}^2}(\lambda_{Ts\sigma}-1)v_{\perp ts\sigma}^2\Big\} = \sum_{j=1}^J\frac{p_{12sj\sigma}}{\omega-c_{sj\sigma}}$,
- $P^u_{s\sigma 21} = \frac{2v_{dsy}}{kv_{ts\sigma}}\Big\{-i\nu_s\frac{k_x}{kv_{ts\sigma}}Z_1 + \frac{k^2k_xv_{\perp ts\sigma}^2}{k^2v_{ts\sigma}^2}Z_2 + \frac{k^2}{kv_{ts\sigma}}v_{dsx}Z_1 + \frac12\frac{k_xk_z^2}{k^2v_{ts\sigma}^2}(\lambda_{Ts\sigma}-1)v_{\perp ts\sigma}^2Z_0\Big\} \simeq 2v_{dsy}\sum_{j=1}^J\frac{b_j}{\omega-c_{sj\sigma}}\Big\{-i\nu_s\frac{k_x}{kv_{ts\sigma}}c_j + \frac{k^2k_xv_{\perp ts\sigma}^2}{k^2v_{ts\sigma}^2}c_j^2 + \frac{k^2}{kv_{ts\sigma}}v_{dsx}c_j + \frac12\frac{k_xk_z^2}{k^2v_{ts\sigma}^2}(\lambda_{Ts\sigma}-1)v_{\perp ts\sigma}^2\Big\} = \sum_{j=1}^J\frac{p_{21sj\sigma}}{\omega-c_{sj\sigma}}$,
- $P^u_{s\sigma 22} = -1 + \frac{2}{kv_{ts\sigma}}\Big\{-i\nu_s\frac12 Z_0 + (\frac12 v_{\perp ts\sigma}^2 + v_{dsy}^2)\frac{k^2}{kv_{ts\sigma}}Z_1\Big\} \simeq -1 + 2\sum_{j=1}^J\frac{b_j}{\omega-c_{sj\sigma}}\Big\{-i\nu_s\frac12 + (\frac12 v_{\perp ts\sigma}^2 + v_{dsy}^2)\frac{k^2}{kv_{ts\sigma}}c_j\Big\} = -1 + \sum_{j=1}^J\frac{p_{22sj\sigma}}{\omega-c_{sj\sigma}}$,
- $P^u_{s\sigma 13} = \frac{2}{kv_{ts\sigma}}\Big\{-i\nu_s\big(\frac{k_xk_zv_{\perp ts\sigma}^2}{k^2v_{ts\sigma}^2}Z_2 + \frac{k_zv_{dsx}}{kv_{ts\sigma}}Z_1 - \frac{k_xk_zv_{\perp ts\sigma}^2}{2k^2v_{ts\sigma}^2}Z_0\big) + \frac{k^2k_xk_zv_{zts}^2v_{\perp ts\sigma}^2}{k^3v_{ts\sigma}^3}Z_3 + \frac{k^2}{kv_{ts\sigma}}[\frac{k_xv_{\perp ts\sigma}^2}{kv_{ts\sigma}}v_{dsz} + v_{dsx}\frac{k_zv_{zts}^2}{kv_{ts\sigma}}]Z_2 + [\frac{k^2}{kv_{ts\sigma}}v_{dsx}v_{dsz} - \frac12\frac{k^2k_xk_zv_{zts}^2v_{\perp ts\sigma}^2}{k^3v_{ts\sigma}^3} - \frac12(\lambda_{Ts\sigma}-1)\frac{k_x^3k_zv_{\perp ts\sigma}^4}{k^3v_{ts\sigma}^3} + \frac12(\lambda_{Ts\sigma}-1)\frac{k_xk_z^3v_{zts}^2v_{\perp ts\sigma}^2}{k^3v_{ts\sigma}^3}]Z_1 + [\frac12(\lambda_{Ts\sigma}-1)\frac{k_xk_z^2v_{\perp ts\sigma}^2}{k^2v_{ts\sigma}^2}v_{dsz} - \frac12(\lambda_{Ts\sigma}-1)v_{dsx}\frac{k_x^2k_zv_{\perp ts\sigma}^2}{k^2v_{ts\sigma}^2}]Z_0\Big\} \simeq 2\sum_{j=1}^J\frac{b_j}{\omega-c_{sj\sigma}}\Big\{-i\nu_s\big(\frac{k_xk_zv_{\perp ts\sigma}^2}{k^2v_{ts\sigma}^2}c_j^2 + \frac{k_zv_{dsx}}{kv_{ts\sigma}}c_j - \frac{k_xk_zv_{\perp ts\sigma}^2}{2k^2v_{ts\sigma}^2}\big) + \frac{k^2k_xk_zv_{zts}^2v_{\perp ts\sigma}^2}{k^3v_{ts\sigma}^3}c_j^3 + \frac{k^2}{kv_{ts\sigma}}[\frac{k_xv_{\perp ts\sigma}^2}{kv_{ts\sigma}}v_{dsz} + v_{dsx}\frac{k_zv_{zts}^2}{kv_{ts\sigma}}]c_j^2 + [\frac{k^2}{kv_{ts\sigma}}v_{dsx}v_{dsz} - \frac12\frac{k^2k_xk_zv_{zts}^2v_{\perp ts\sigma}^2}{k^3v_{ts\sigma}^3} - \frac12(\lambda_{Ts\sigma}-1)\frac{k_x^3k_zv_{\perp ts\sigma}^4}{k^3v_{ts\sigma}^3} + \frac12(\lambda_{Ts\sigma}-1)\frac{k_xk_z^3v_{zts}^2v_{\perp ts\sigma}^2}{k^3v_{ts\sigma}^3}]c_j + [\frac12(\lambda_{Ts\sigma}-1)\frac{k_xk_z^2v_{\perp ts\sigma}^2}{k^2v_{ts\sigma}^2}v_{dsz} - \frac12(\lambda_{Ts\sigma}-1)v_{dsx}\frac{k_x^2k_zv_{\perp ts\sigma}^2}{k^2v_{ts\sigma}^2}]\Big\} = \sum_{j=1}^J\frac{p_{13sj\sigma}}{\omega-c_{sj\sigma}}$,
- $P^u_{s\sigma 31} = \frac{2}{kv_{ts\sigma}}\Big\{-i\nu_s\big(\frac{k_xk_zv_{zts}^2}{k^2v_{ts\sigma}^2}Z_2 + \frac{k_xv_{dsz}}{kv_{ts\sigma}}Z_1 - \frac{k_xk_zv_{zts}^2}{2k^2v_{ts\sigma}^2}Z_0\big) + \frac{k^2k_xk_zv_{zts}^2v_{\perp ts\sigma}^2}{k^3v_{ts\sigma}^3}Z_3 + \frac{k^2}{kv_{ts\sigma}}[\frac{k_xv_{\perp ts\sigma}^2}{kv_{ts\sigma}}v_{dsz} + v_{dsx}\frac{k_zv_{zts}^2}{kv_{ts\sigma}}]Z_2 + [\frac{k^2}{kv_{ts\sigma}}v_{dsx}v_{dsz} - \frac12\frac{k^2k_xk_zv_{zts}^2v_{\perp ts\sigma}^2}{k^3v_{ts\sigma}^3} - \frac12(\lambda_{Ts\sigma}-1)\frac{k_x^3k_zv_{\perp ts\sigma}^4}{k^3v_{ts\sigma}^3} + \frac12(\lambda_{Ts\sigma}-1)\frac{k_xk_z^3v_{zts}^2v_{\perp ts\sigma}^2}{k^3v_{ts\sigma}^3}]Z_1 + [\frac12(\lambda_{Ts\sigma}-1)\frac{k_xk_z^2v_{\perp ts\sigma}^2}{k^2v_{ts\sigma}^2}v_{dsz} - \frac12(\lambda_{Ts\sigma}-1)v_{dsx}\frac{k_x^2k_zv_{\perp ts\sigma}^2}{k^2v_{ts\sigma}^2}]Z_0\Big\} \simeq 2\sum_{j=1}^J\frac{b_j}{\omega-c_{sj\sigma}}\Big\{-i\nu_s\big(\frac{k_xk_zv_{zts}^2}{k^2v_{ts\sigma}^2}c_j^2 + \frac{k_xv_{dsz}}{kv_{ts\sigma}}c_j - \frac{k_xk_zv_{zts}^2}{2k^2v_{ts\sigma}^2}\big) + \frac{k^2k_xk_zv_{zts}^2v_{\perp ts\sigma}^2}{k^3v_{ts\sigma}^3}c_j^3 + \frac{k^2}{kv_{ts\sigma}}[\frac{k_xv_{\perp ts\sigma}^2}{kv_{ts\sigma}}v_{dsz} + v_{dsx}\frac{k_zv_{zts}^2}{kv_{ts\sigma}}]c_j^2 + [\frac{k^2}{kv_{ts\sigma}}v_{dsx}v_{dsz} - \frac12\frac{k^2k_xk_zv_{zts}^2v_{\perp ts\sigma}^2}{k^3v_{ts\sigma}^3} - \frac12(\lambda_{Ts\sigma}-1)\frac{k_x^3k_zv_{\perp ts\sigma}^4}{k^3v_{ts\sigma}^3} + \frac12(\lambda_{Ts\sigma}-1)\frac{k_xk_z^3v_{zts}^2v_{\perp ts\sigma}^2}{k^3v_{ts\sigma}^3}]c_j + [\frac12(\lambda_{Ts\sigma}-1)\frac{k_xk_z^2v_{\perp ts\sigma}^2}{k^2v_{ts\sigma}^2}v_{dsz} - \frac12(\lambda_{Ts\sigma}-1)v_{dsx}\frac{k_x^2k_zv_{\perp ts\sigma}^2}{k^2v_{ts\sigma}^2}]\Big\} = \sum_{j=1}^J\frac{p_{31sj\sigma}}{\omega-c_{sj\sigma}}$,
- $P^u_{s\sigma 23} = \frac{2v_{dsy}}{kv_{ts\sigma}}\Big\{-i\nu_s\frac{k_z}{kv_{ts\sigma}}Z_1 + \frac{k^2k_zv_{zts}^2}{k^2v_{ts\sigma}^2}Z_2 + \frac{k^2}{kv_{ts\sigma}}v_{dsz}Z_1 - \frac12(\lambda_{Ts\sigma}-1)\frac{k_x^2k_zv_{\perp ts\sigma}^2}{k^2v_{ts\sigma}^2}Z_0\Big\} \simeq 2v_{dsy}\sum_{j=1}^J\frac{b_j}{\omega-c_{sj\sigma}}\Big\{-i\nu_s\frac{k_z}{kv_{ts\sigma}}c_j + \frac{k^2k_zv_{zts}^2}{k^2v_{ts\sigma}^2}c_j^2 + \frac{k^2}{kv_{ts\sigma}}v_{dsz}c_j - \frac12(\lambda_{Ts\sigma}-1)\frac{k_x^2k_zv_{\perp ts\sigma}^2}{k^2v_{ts\sigma}^2}\Big\} = \sum_{j=1}^J\frac{p_{23sj\sigma}}{\omega-c_{sj\sigma}}$,
- $P^u_{s\sigma 32} = \frac{2v_{dsy}}{kv_{ts\sigma}}\Big\{\frac{k^2k_zv_{zts}^2}{k^2v_{ts\sigma}^2}Z_2 + \frac{k^2}{kv_{ts\sigma}}v_{dsz}Z_1 - \frac12(\lambda_{Ts\sigma}-1)\frac{k_x^2k_zv_{\perp ts\sigma}^2}{k^2v_{ts\sigma}^2}Z_0\Big\} \simeq 2v_{dsy}\sum_{j=1}^J\frac{b_j}{\omega-c_{sj\sigma}}\Big\{\frac{k^2k_zv_{zts}^2}{k^2v_{ts\sigma}^2}c_j^2 + \frac{k^2}{kv_{ts\sigma}}v_{dsz}c_j - \frac12(\lambda_{Ts\sigma}-1)\frac{k_x^2k_zv_{\perp ts\sigma}^2}{k^2v_{ts\sigma}^2}\Big\} = \sum_{j=1}^J\frac{p_{32sj\sigma}}{\omega-c_{sj\sigma}}$,
- $P^u_{s\sigma 33} = -1 + \frac{2}{kv_{ts\sigma}}\Big\{-i\nu_s\big(\frac{k_z^2v_{zts}^2}{k^2v_{ts\sigma}^2}Z_2 + \frac{k_zv_{dsz}}{kv_{ts\sigma}}Z_1 + \frac{k_x^2v_{\perp ts\sigma}^2}{2k^2v_{ts\sigma}^2}Z_0\big) + \frac{k^2k_z^2v_{zts}^4}{k^3v_{ts\sigma}^3}Z_3 + 2\frac{k^2k_zv_{zts}^2}{k^2v_{ts\sigma}^2}v_{dsz}Z_2 + [\frac12\frac{k^2k_x^2v_{zts}^2v_{\perp ts\sigma}^2}{k^3v_{ts\sigma}^3} + \frac{k^2}{kv_{ts\sigma}}v_{dsz}^2 - (\lambda_{Ts\sigma}-1)\frac{k_x^2k_z^2v_{zts}^2v_{\perp ts\sigma}^2}{k^3v_{ts\sigma}^3}]Z_1 - (\lambda_{Ts\sigma}-1)\frac{k_x^2k_zv_{\perp ts\sigma}^2}{k^2v_{ts\sigma}^2}v_{dsz}Z_0\Big\} \simeq -1 + 2\sum_{j=1}^J\frac{b_j}{\omega-c_{sj\sigma}}\Big\{-i\nu_s\big(\frac{k_z^2v_{zts}^2}{k^2v_{ts\sigma}^2}c_j^2 + \frac{k_zv_{dsz}}{kv_{ts\sigma}}c_j + \frac{k_x^2v_{\perp ts\sigma}^2}{2k^2v_{ts\sigma}^2}\big) + \frac{k^2k_z^2v_{zts}^4}{k^3v_{ts\sigma}^3}c_j^3 + 2\frac{k^2k_zv_{zts}^2}{k^2v_{ts\sigma}^2}v_{dsz}c_j^2 + [\frac12\frac{k^2k_x^2v_{zts}^2v_{\perp ts\sigma}^2}{k^3v_{ts\sigma}^3} + \frac{k^2}{kv_{ts\sigma}}v_{dsz}^2 - (\lambda_{Ts\sigma}-1)\frac{k_x^2k_z^2v_{zts}^2v_{\perp ts\sigma}^2}{k^3v_{ts\sigma}^3}]c_j - (\lambda_{Ts\sigma}-1)\frac{k_x^2k_zv_{\perp ts\sigma}^2}{k^2v_{ts\sigma}^2}v_{dsz}\Big\} = -1 + \sum_{j=1}^J\frac{p_{33sj\sigma}}{\omega-c_{sj\sigma}}$.

In the above, for example, $p_{11sj\sigma} = 2b_j\Big\{-i\nu_s\big(\frac{k_x^2v_{\perp ts\sigma}^2}{k^2v_{ts\sigma}^2}c_j^2 + \frac{k_xv_{dsx}}{kv_{ts\sigma}}c_j + \frac{k_z^2v_{zts}^2}{2k^2v_{ts\sigma}^2}\big) + \frac{k^2k_x^2v_{\perp ts\sigma}^4}{k^3v_{ts\sigma}^3}c_j^3 + 2\frac{k^2k_xv_{\perp ts\sigma}^2}{k^2v_{ts\sigma}^2}v_{dsx}c_j^2 + [\frac12 v_{\perp ts\sigma}^2\frac{k^2k_z^2v_{zts}^2}{k^3v_{ts\sigma}^3} + \frac{k^2}{kv_{ts\sigma}}v_{dsx}^2 + (\lambda_{Ts\sigma}-1)\frac{k_x^2k_z^2v_{\perp ts\sigma}^4}{k^3v_{ts\sigma}^3}]c_j + \frac{k_xk_z^2}{k^2v_{ts\sigma}^2}(\lambda_{Ts\sigma}-1)v_{\perp ts\sigma}^2v_{dsx}\Big\}$, and others are similar and thus we have not written them out explicitly.

We find the result is very simple by use $Z_{0,1,2,3}$, which would also make the EM3D-M case be much simpler than the previous BO-K [7] derivation. This is also why we use $Z_{0,1}$ in the ES3D case, which gives a more compact form.

Thus, we obtain the relations between $\boldsymbol{J}^u$ and $\boldsymbol{E}$, which has the following form (with $\sum_{s=u}$)

$$\begin{pmatrix} J_x^u \\ J_y^u \\ J_z^u \end{pmatrix} = -i\epsilon_0\begin{pmatrix} \frac{b_{11}^u}{\omega} + \sum_{sj\sigma}\frac{b_{sj\sigma 11}}{\omega - c_{sj\sigma}} & \frac{b_{12}^u}{\omega} + \sum_{sj\sigma}\frac{b_{sj\sigma 12}}{\omega - c_{sj\sigma}} & \frac{b_{13}^u}{\omega} + \sum_{sj\sigma}\frac{b_{sj\sigma 13}}{\omega - c_{sj\sigma}} \\ \frac{b_{21}^u}{\omega} + \sum_{sj\sigma}\frac{b_{sj\sigma 21}}{\omega - c_{sj\sigma}} & \frac{b_{22}^u}{\omega} + \sum_{sj\sigma}\frac{b_{sj\sigma 22}}{\omega - c_{sj\sigma}} & \frac{b_{23}^u}{\omega} + \sum_{sj\sigma}\frac{b_{sj\sigma 23}}{\omega - c_{sj\sigma}} \\ \frac{b_{31}^u}{\omega} + \sum_{sj\sigma}\frac{b_{sj\sigma 31}}{\omega - c_{sj\sigma}} & \frac{b_{32}^u}{\omega} + \sum_{sj\sigma}\frac{b_{sj\sigma 32}}{\omega - c_{sj\sigma}} & \frac{b_{33}^u}{\omega} + \sum_{sj\sigma}\frac{b_{sj\sigma 33}}{\omega - c_{sj\sigma}} \end{pmatrix}\begin{pmatrix} E_x \\ E_y \\ E_z \end{pmatrix}, \tag{127}$$

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# 3. Obtain all solutions of kinetic dispersion relation

## 3.3 With ring beam, perpendicular drifts, collision, etc (Xie2019CPC)



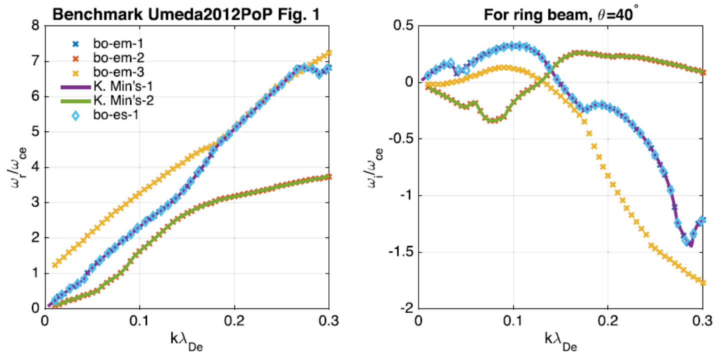


**Fig. 1.** Benchmark the electromagnetic ring beam dispersion relation between BO-K and Min's code [12] with the parameters in Ref. [13] Fig. 1 for $\theta = 40°$, which shows very good agreement. BO-K gives all the three unstable branches, whereas the third branch 'bo-em-3' could easily be missed by the iterative root finding approaches used in Min's [12] or Umeda's [13] codes. One branch of the electrostatic BO-K solution is also shown for reference, which is close to the electromagnetic one implies that this branch is essentially an electrostatic mode. The results also agree well with the PIC simulation result in Ref. [13].

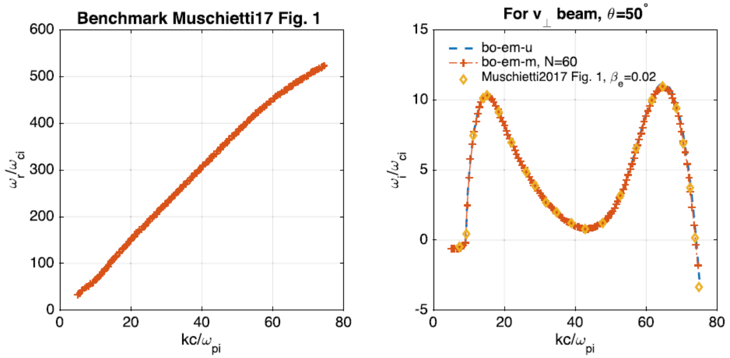


**Fig. 2.** Benchmark the electromagnetic perpendicular beam dispersion relation between BO-K and Ref. [5] Fig. 1 with the parameters $\theta = 50°$ and $\beta_e = 0.02$, which shows very good agreement. Ref. [5] assumes unmagnetized ions and magnetized electron. 'bo-em-u' uses the same assumption; whereas 'bo-em-m' uses also magnetized ions with $N = 60$ where the $\sum_n$ has convergent. The agreement between '-U' and '-M' versions also implies that the unmagnetized ions is a valid assumption for this case.

To date, this is the most comprehensive Maxwellian-based dispersion relation solver.

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### BO - A powerful fluid and kinetic plasma wave and instability analysis tool

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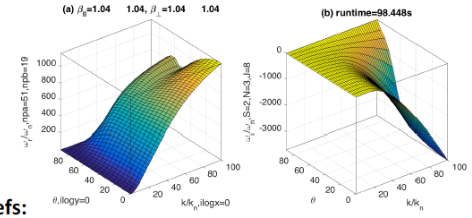
CCF-ENN, 2019-10-06

**-Introduction**

- BO ('波', i.e., 'wave' in Chinese) currently includes BO-F (PDRF, multi-fluid solver) and BO-K (PDRK, kinetic solver).
- BO-K (PDRK) solves uniform plasma dispersion relation with an extended Maxwellian based equilibrium distribution function.
- For D(omega, k)=0, give k, solve series omega(s).
- ***What make BO attractive?*** Solves the difficulty of root finding, i.e., the *first solver* not requires initial guess and can *give all the important solutions at one time*. You *do not need luck* any more!
- ***Supports:*** anisotropic temperature / loss cone / *drift in arbitrary direction / ring beam / collision*, unmagnetized / magnetized species, electrostatic/electromagnetic/Darwin, can kpara<=0, etc.

**-Typical example**

- Fast-magnetosonic/whistler dispersion surface




**- Refs:**

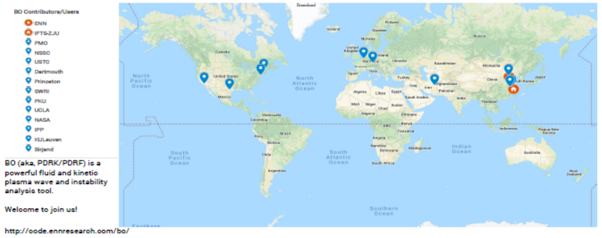

- H. S. Xie & Y. Xiao, Plasma Sci. & Tech., 18, 2, 97 (2016).
- H. S. Xie, BO: A unified tool for plasma waves and instabilities analysis, Comp. Phys. Comm., 244, 343 (2019).
- H. S. Xie, Comp. Phys. Comm., 185, 670 (2014).


**-Solvers compare***

| | BO (PDRK) [Xie16,19] | WHAMP [Ronnmark82] | KUPDAP [Sugiyama15] | DSHARK [Astfalk15] |
|---|---|---|---|---|
| Initial guess? | Not required | Must | Must | Must |
| Fast? | Middle | Fast | Middle | Middle |
| Support high harmonic? | Easy | Difficult | Difficult | Difficult |
| Separate modes? | Easy | Difficult | Difficult | Difficult |
| All solutions? | Yes | No | Multi solutions in given range | No |
| With perp drift? | Yes | No | No | No |
| With collision? | Yes | No | No | No |
| Key feature | All solutions, rich models | Fast, widely used | Multi solutions | Kappa distribution |

*Other solvers not shown here, e.g., P. Gary's, Verscharen's, …

**BO Contributors/Users Map (2018)**



**-Download:**

- http://code.ennresearch.com/bo/ (with latest version)
- https://github.com/hsxie/pdrk (v181027 and older)

**Welcome to join us/spread this poster!**

**- Ackn:**

- R. Denton, Xin Tao, Jin-song Zhao, Zhong-wei Yang, Chao-jie Zhang, Can Huang, Wen-ya Li, Liang Wang, Kyungguk Min, Yang Li, Kyunghwan Dokgo, J. L. Burch,…

Feature: Fast, accurate, obtain all solutions (except less important strong damped modes) of kinetic EM3D/ES3D/Darwin & fluid systems. Without singularity, no need initial guess for root finding.

https://github.com/hsxie/bo

# 3. Obtain all solutions of kinetic dispersion relation

## 3.4 Arbitrary distributions (Xie2025)

$$\bar{D}(\omega,\boldsymbol{k}) = \left|\boldsymbol{K}(\omega,\boldsymbol{k}) + (\boldsymbol{kk} - k^2\boldsymbol{I})\frac{c^2}{\omega^2}\right| = 0, \qquad (1)$$

where $\boldsymbol{K} = \boldsymbol{I} + \boldsymbol{Q} = \boldsymbol{I} - \frac{\boldsymbol{\sigma}}{i\omega\epsilon_0}$, $\boldsymbol{Q} = -\frac{\boldsymbol{\sigma}}{i\omega\epsilon_0}$, and

$$\boldsymbol{\sigma} = -i\sum_s \frac{q_s^2 n_{s0}}{m_s}\sum_{n=-\infty}^{\infty}\int_{-\infty}^{\infty}\int_0^{\infty}\frac{2\pi v_\perp dv_\perp dv_\parallel}{\omega - n\omega_{cs} - k_\parallel v_\parallel}\boldsymbol{\Pi}_s, (2)$$

with

$$\boldsymbol{\Pi}_s = \begin{bmatrix} A_s\frac{n^2 v_\perp}{\mu_s^2}J_n^2 & iA_s\frac{n v_\perp}{\mu_s}J_nJ_n' & B_s\frac{n v_\perp}{\mu_s}J_n^2 \\ -iA_s\frac{n v_\perp}{\mu_s}J_nJ_n' & A_s v_\perp J_n'^2 & -iB_s v_\perp J_nJ_n' \\ A_s\frac{n v_\parallel}{\mu_s}J_n^2 & iA_s v_\parallel J_nJ_n' & B_s v_\parallel J_n^2 \end{bmatrix}, (3)$$

where $\mu_s = \frac{k_\perp v_\perp}{\omega_{cs}}$, $A_s = \left(1 - \frac{k_\parallel v_\parallel}{\omega}\right)\frac{\partial f_{s0}}{\partial v_\perp} + \frac{k_\parallel v_\perp}{\omega}\frac{\partial f_{s0}}{\partial v_\parallel}$, and $B_s = \frac{n\omega_{cs}v_\parallel}{\omega v_\perp}\frac{\partial f_{s0}}{\partial v_\perp} + \left(1 - \frac{n\omega_{cs}}{\omega}\right)\frac{\partial f_{s0}}{\partial v_\parallel}$. Eq. (2) is valid for non-relativistic, arbitrary gyrotropic distributions. Here,

In this work, we write the normalized distribution function of species $s$ to orthogonal basis functions expansion form

$$f_{s0}(v_\parallel, v_\perp) = \sum_{l=-\infty}^{\infty}\sum_{m=-\infty}^{\infty} a_{s0,lm}W_{sz}(v_\parallel)W_{sx}(v_\perp)\rho_{sz,l}(v_\parallel)u_{sx,m}(v_\perp), \quad -\infty < v_\parallel < \infty, \quad 0 \le v_\perp < \infty, \qquad (5)$$

where $\rho_{sz,l}$ and $u_{sx,m}$ are basis functions for parallel and perpendicular directions, with $W_{sz}$ and $W_{sx}$ be corresponding weight functions, which satisfy

$$\int_{-\infty}^{\infty} W_{sz}(v)\rho_n(v)\rho_{n'}^*(v)dv = A_{sz}\delta_{n,n'}, \quad \int_{-\infty}^{\infty} W_{sx}(v)u_n(v)u_{n'}^*(v)dv = A_{sx}\delta_{n,n'}. \qquad (6)$$

Thus, we have

$$a_{s0,lm} = \frac{1}{A_{sx}A_{sz}}\int_{-\infty}^{\infty}\int_{-\infty}^{\infty} f_{s0}(v_\parallel, v_\perp)\rho_{sz,l}^*((v_\parallel)h_{sx,m}^*(v_\perp)dv_\parallel dv_\perp. \qquad (7)$$

Note also that $\int f_{s0}dv^3 = \int_{-\infty}^{\infty}\int_0^{\infty} 2\pi v_\perp f_{s0}dv_\perp dv_\parallel = 1$.

| Distribution $f_0(v_\parallel, v_\perp)$ | Drift bi-Maxwellian Ring Beam | Drift bi-Kappa | Drift product bi-kappa | Shell |
|---|---|---|---|---|
| Form ($v_\parallel = v_z$, $v_\perp = \sqrt{v_x^2 + v_y^2}$, $v = \sqrt{v_\parallel^2 + v_\perp^2}$) | $c_0\exp\left[-\frac{(v_\parallel - v_{dz})^2}{v_{tz}^2} - \frac{(v_\perp - v_{dr})^2}{v_{tp}^2}\right]$ | $c_0\left[1 + \frac{(v_\parallel - v_{dz})^2}{\kappa v_{tz}^2} + \frac{v_\perp^2}{\kappa v_{tp}^2}\right]^{-\kappa-1}$ | $c_0\left[1 + \frac{(v_\parallel - v_{dz})^2}{\kappa_\parallel v_{tz}^2}\right]^{-\kappa_\parallel-1}\left[1 + \frac{v_\perp^2}{\kappa_\perp v_{tp}^2}\right]^{-\kappa_\perp-1}$ | $c_0\exp\left[-\frac{(v - v_d)^2}{v_t^2}\right]$ |
| Coefficient | $c_0 = \frac{1}{\pi^{\frac{3}{2}}v_{tz}v_{tp}^2A}$, $A = e^{-\frac{v_{dr}^2}{v_{tp}^2}} + \frac{\sqrt{\pi}v_{dr}}{v_{tp}}\mathrm{erfc}\left(-\frac{v_{dr}}{v_{tp}}\right)$, $v_{tz,p} = \sqrt{\frac{2k_BT_{z,p}}{m}}$ | $c_0 = \frac{1}{\pi^{\frac{3}{2}}v_{tz}v_{tp}^2A}$, $A = \frac{\Gamma(\kappa+1)}{\kappa^{3/2}\Gamma\left(\kappa - \frac{1}{2}\right)}$, $v_{tz,p} = \sqrt{\frac{2k_BT_{z,p}}{m}(1 - \frac{3}{2\kappa})}$ | $c_0 = \frac{1}{\pi^{\frac{3}{2}}v_{tz}v_{tp}^2A}$, $A = \frac{\Gamma(\kappa_\parallel+1)}{\kappa_\parallel^{1/2}\Gamma\left(\kappa_\parallel + \frac{1}{2}\right)}$, $v_{tz} = \sqrt{\frac{2k_BT_z}{m}(1 - \frac{1}{2\kappa_\parallel})}$, $v_{tp} = \sqrt{\frac{2k_BT_p}{m}(1 - \frac{1}{\kappa_\perp})}$ | $c_0 = \frac{1}{\pi^{\frac{3}{2}}v_t^3A}$, $A = \frac{2}{\sqrt{\pi}}\frac{v_{dr}}{v_{tp}}e^{-\frac{v_{dr}^2}{v_{tp}^2}} + \left(2\frac{v_{dr}^2}{v_{tp}^2} + 1\right)\mathrm{erfc}\left(-\frac{v_{dr}}{v_{tp}}\right)$ |

FIG. 1: Typical non-Maxwellian velocity distributions met in space and laboratory plasmas.

Can we obtain all solutions for arbitrary distribution fastly & accurately? Yes, and the computation time not beyond too much to Maxwellian case!

to the Maxwellian distribution. The Hermite basis yields a form suitable for J-pole expansion and facilitates a matrix approach [9, 10] with a smaller matrix dimension, making it possible to obtain all solutions without requiring an initial guess. In this work, we demonstrate results using the Hermite-Hermite (HH) basis for both parallel and perpendicular directions. Detailed derivations of the equations, along with results using GPDF-Hermite (GH) and GPDF-GPDF (GG) bases, are provided in the supplementary material.

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# 3. Obtain all solutions of kinetic dispersion relation

## 3.4 Arbitrary distributions (Xie2025)

For the HH expansion, we use the following form:

$$f_{s0}(v_\parallel, v_\perp) = c_{s0} \sum_{l=0}^{\infty} \sum_{m=0}^{\infty} a_{s,lm} \cdot g_{sz,l}(v_\parallel) \cdot g_{sx,m}(v_\perp), \quad (4)$$

where $g_{sz,l}(v_\parallel) = \left(\frac{v_\parallel - d_{sz}}{L_{sz}}\right)^l e^{-\left(\frac{v_\parallel - d_{sz}}{L_{sz}}\right)^2}$, $g_{sx,m}(v_\perp) = \left(\frac{v_\perp - d_{sx}}{L_{sx}}\right)^m e^{-\left(\frac{v_\perp - d_{sx}}{L_{sx}}\right)^2}$. Here, $L_{sz}$ and $L_{sx}$ are the velocity widths, while $d_{sz}$ and $d_{sx}$ are the drift velocities. The normalization coefficient is given by $c_{s0} = \frac{1}{\pi^{3/2} L_{sz} L_{sx}^2 R_s}$, where $R_s = \exp\left(-\frac{d_{sx}^2}{L_{sx}^2}\right) + \frac{\sqrt{\pi} d_{sx}}{L_{sx}} \mathrm{erfc}\left(-\frac{d_{sx}}{L_{sx}}\right)$, and $\mathrm{erfc}(-x) = 1 - \mathrm{erf}(-x) = 1 + \mathrm{erf}(x)$ is the complementary error function. The coefficients $a_{s,lm}$ can be readily calculated using orthogonal Hermite basis functions.

The benefits of this choice of basis functions include: (1) It naturally reduces to the drift bi-Maxwellian ring-beam case [10] by retaining only the lowest-order term with $a_{s,00} = 1 \neq 0$. (2) The parallel integral can be simplified to involve only a single Maxwellian $Z$ function, and the perpendicular integral can also be reduced to a single term initially. The second benefit allows the J-pole matrix method from BO [10] to be directly applied here, with the matrix dimension remaining unchanged, though

We define $f_{s0,lm}(v_\parallel, v_\perp) \equiv a_{s,lm} g_{sz,l}(v_\parallel) g_{sx,m}(v_\perp)$, $f_{s0z,l}(v_\parallel) \equiv g_{sz,l}(v_\parallel)$, $f_{s0x,m}(v_\perp) \equiv g_{sx,m}(v_\perp)$. Hence, we have $\frac{\partial f_{s0z,l}(v_\parallel)}{\partial v_\parallel} = -\frac{1}{L_{sz}}\Big[2 f_{s0z,l+1} - l f_{s0z,l-1}\Big]$, $\frac{\partial f_{s0x,m}(v_\perp)}{\partial v_\perp} = -\frac{1}{L_{sx}}\Big[2 f_{s0x,m+1} - m f_{s0x,m-1}\Big]$. Here, we set $f_{s0z,l} = f_{s0x,m} = 0$ for all $l, m < 0$. The above derivatives equations are valid for all $l, m = 0, 1, 2, \cdots$.

For parallel integral $[\mathrm{Im}(\zeta_{sn}) > 0]$, we define

$$Z_{l,p}(\zeta_{sn}) \equiv -\frac{k_\parallel}{L_{sz}^p \sqrt{\pi}} \int_{-\infty}^{\infty} \frac{v_\parallel^p (\frac{v_\parallel - d_{sz}}{L_{sz}})^l e^{-(\frac{v_\parallel - d_{sz}}{L_{sz}})^2}}{\omega - k_\parallel v_\parallel - n\omega_{cs}} dv_\parallel$$
$$= \frac{1}{\sqrt{\pi}} \int_{-\infty}^{\infty} \frac{g_{l,p}(z)}{z - \zeta_{sn}} dz, \quad g_{l,p}(z) \equiv (z+d)^p z^l e^{-z^2}. \quad (5)$$

We have $Z_{0,0}(\zeta) = Z(\zeta)$ and redifine $Z_{l,0} \equiv Z_l$ and $I_l \equiv \pi^{-1/2} \int_{-\infty}^{\infty} x^l e^{-x^2} dx$, with $\zeta_{sn} = \frac{\omega - k_\parallel d_{sz} - n\omega_{cs}}{k_\parallel L_{sz}}$, $d = \frac{d_{sz}}{L_{sz}}$, $p = 0, 1, 2$. These plasma dispersion functions can be computed efficiently to high accuracy and are also analytically continuous for $\mathrm{Im}(\zeta_{sn}) \leq 0$ [18]. For the perpendicular integral, we define

$$\Gamma_{\{a,b,c\}n,m,p}(a_s, d_s) \equiv \frac{1}{L_{sx}^{p+1}} \int_0^\infty v_\perp^p \{J_n^2(\frac{k_\perp v_\perp}{\omega_{cs}}), J_n(\frac{k_\perp v_\perp}{\omega_{cs}}) J_n'(\frac{k_\perp v_\perp}{\omega_{cs}}),$$
$$J_n'^2(\frac{k_\perp v_\perp}{\omega_{cs}})\}(\frac{v_\perp - d_{sx}}{L_{sx}})^m e^{-(\frac{v_\perp - d_{sx}}{L_{sx}})^2} dv_\perp$$
$$= \int_0^\infty x^p \{J_n^2(a_s x), J_n(a_s x) J_n'(a_s x), J_n'^2(a_s x)\}(x - d_s)^m e^{-(x-d_s)^2} dx, \quad (6)$$

with $a_s = k_\perp \rho_{cs}$, $\rho_{cs} = L_{sx}/\omega_{cs}$, $d_s = d_{sx}/L_{sx}$, $p = 0, 1, 2, 3$. These integrals can be computed numerically [10].

For the Maxwellian $Z$ function J-pole expansion, we have[10]

$$Z_l(\zeta) \simeq \sum_{j=1}^{J} \frac{b_j c_j^l}{\zeta - c_j}. \quad (7)$$

Here, we have used[23] $\sum_j b_j = -1$, $\sum_j b_j c_j = 0$, $\sum_j b_j c_j^2 = -1/2$, $\sum_j b_j c_j^3 = 0$, $\cdots$. This also means that if we need a higher-order Hermite fitting of the distribution function, the corresponding coefficients should also be kept to a similar order. Typically, we need to set $J \geq l_{\max} + 4$. To ensure double precision, we have calculated up to $J = 24$. Hence, the present solver can support $l_{\max} \simeq 20$. While higher values of $l_{\max}$ can also be computed, the accuracy would decrease.

To seek an equivalent linear system, Maxwell's equations are

$$\partial_t \boldsymbol{E} = c^2 \nabla \times \boldsymbol{B} - \boldsymbol{J}/\epsilon_0, \quad (8a)$$
$$\partial_t \boldsymbol{B} = -\nabla \times \boldsymbol{E}, \quad (8b)$$

which do not need to be changed. We only need to seek a new linear system for $\boldsymbol{J} = \boldsymbol{\sigma} \cdot \boldsymbol{E}$.

Key:
- Rational form of Z
- Naturally analytically continuous for damped modes

Note: GPDF basis matrix similar to kappa distribution (Bai23PoP, 25CPC), complicated than Hermite basis

Straightforward derivations, step by step, for Hermite-Hermite (J-pole matrix), GPDF-Hermite (accurate), GPDF-GPDF (can use FFT) basis, etc

# 3. Obtain all solutions of kinetic dispersion relation

## 3.4 Arbitrary distributions (Xie2025)

Define

$$\boldsymbol{X}_{sn} = \int_{-\infty}^{\infty}\int_{0}^{\infty} \frac{2\pi\omega v_\perp dv_\perp dv_\parallel}{\omega - k_\parallel v_\parallel - n\omega_{cs}} \times \begin{bmatrix} A_s\frac{n^2 v_\perp}{\mu_s^2}J_n^2 & iA_s\frac{n v_\perp}{\mu_s}J_nJ_n' & B_s\frac{n v_\perp}{\mu_s}J_n^2 \\ -iA_s\frac{n v_\perp}{\mu_s}J_nJ_n' & A_s v_\perp J_n'^2 & -iB_s v_\perp J_nJ_n' \\ A_s\frac{n v_\parallel}{\mu_s}J_n^2 & iA_s v_\parallel J_nJ_n' & B_s v_\parallel J_n^2 \end{bmatrix}. \quad (31)$$

We have

$$\begin{aligned} X_{sn11} &= \int_{-\infty}^{\infty}\int_{0}^{\infty} \frac{2\pi\omega v_\perp dv_\perp dv_\parallel}{\omega - k_\parallel v_\parallel - n\omega_{cs}} \times A_s\frac{n^2 v_\perp}{\mu_s^2}J_n^2 = \frac{n^2\omega_{cs}^2}{k_\perp^2}\int_{-\infty}^{\infty}\int_{0}^{\infty} \frac{2\pi\omega dv_\perp dv_\parallel}{\omega - k_\parallel v_\parallel - n\omega_{cs}} \times A_sJ_n^2 \\ &= -c_{s0}\frac{n^2\omega_{cs}^2}{k_\perp^2}\sum_l\sum_m a_{s,lm}\int_{-\infty}^{\infty}\int_{0}^{\infty} \frac{2\pi\omega dv_\perp dv_\parallel}{\omega - k_\parallel v_\parallel - n\omega_{cs}} \times \Big\{\frac{\omega - k_\parallel v_\parallel}{\omega} f_{s0z,l}\frac{1}{L_{sx}}[2f_{s0x,m+1} - mf_{s0x,m-1}] \\ &+ \frac{k_\parallel v_\perp}{\omega} f_{s0x,m}\frac{1}{L_{sz}}[2f_{s0z,l+1} - lf_{s0z,l-1}]\Big\}J_n^2 \\ &= 2\frac{1}{L_{sz}L_{sx}^2R_s}\frac{n^2\omega_{cs}^2}{k_\perp^2}\sum_l\sum_m a_{s,lm}\Big\{(\frac{\omega}{k_\parallel}Z_{l,0} - L_{sz}Z_{l,1})[2\Gamma_{an,m+1,0} - m\Gamma_{an,m-1,0}] + \Gamma_{an,m,1}\frac{L_{sx}^2}{L_{sz}}[2Z_{l+1,0} - lZ_{l-1,0}]\Big\} \\ &= \frac{2}{R_s}\frac{n^2\omega_{cs}^2}{k_\perp^2L_{sx}^2}\sum_l\sum_m a_{s,lm}\Big\{(\zeta_{sn}Z_{l,0} + \frac{n\omega_{cs}}{k_\parallel L_{sz}}Z_{l,0} - Z_{l+1,0})[2\Gamma_{an,m+1,0} - m\Gamma_{an,m-1,0}] + \Gamma_{an,m,1}\frac{L_{sx}^2}{L_{sz}^2}[2Z_{l+1,0} - lZ_{l-1,0}]\Big\} \\ &= \frac{2}{R_s}\frac{n^2\omega_{cs}^2}{k_\perp^2L_{sx}^2}\sum_l\sum_m a_{s,lm}\Big\{(\frac{n\omega_{cs}}{k_\parallel L_{sz}}Z_l - I_l)(2\Gamma_{an,m+1,0} - m\Gamma_{an,m-1,0}) + \Gamma_{an,m,1}\frac{L_{sx}^2}{L_{sz}^2}(2Z_{l+1} - lZ_{l-1})\Big\}, \end{aligned}$$

It is thus easy to find that after the J-pole expansion, the relations between $\boldsymbol{J}$ and $\boldsymbol{E}$ have the following form

$$\frac{\boldsymbol{\sigma}}{-i\epsilon_0} = \begin{pmatrix} \frac{b_{11}}{\omega} + \sum_{snj}\frac{b_{snj11}}{\omega - c_{snj}} & \frac{b_{12}}{\omega} + \sum_{snj}\frac{b_{snj12}}{\omega - c_{snj}} & \frac{b_{13}}{\omega} + \sum_{snj}\frac{b_{snj13}}{\omega - c_{snj}} \\ \frac{b_{21}}{\omega} + \sum_{snj}\frac{b_{snj21}}{\omega - c_{snj}} & \frac{b_{22}}{\omega} + \sum_{snj}\frac{b_{snj22}}{\omega - c_{snj}} & \frac{b_{23}}{\omega} + \sum_{snj}\frac{b_{snj23}}{\omega - c_{snj}} \\ \frac{b_{31}}{\omega} + \sum_{snj}\frac{b_{snj31}}{\omega - c_{snj}} & \frac{b_{32}}{\omega} + \sum_{snj}\frac{b_{snj32}}{\omega - c_{snj}} & \frac{b_{33}}{\omega} + \sum_{snj}\frac{b_{snj33}}{\omega - c_{snj}} \end{pmatrix} \quad (9)$$

Combining Eqs. (8) and (9), the equivalent linear system for the electromagnetic dispersion relation can be obtained as

$$\left\{\begin{aligned} \omega v_{snjx} &= c_{snj}v_{snjx} + b_{snj11}E_x + b_{snj12}E_y + b_{snj13}E_z, \\ \omega j_x &= b_{11}E_x + b_{12}E_y + b_{13}E_z, \\ iJ_x\epsilon_0 &= j_x + \textstyle\sum_{snj} v_{snjx}, \\ \omega v_{snjy} &= c_{snj}v_{snjy} + b_{snj21}E_x + b_{snj22}E_y + b_{snj23}E_z, \\ \omega j_y &= b_{21}E_x + b_{22}E_y + b_{23}E_z, \\ iJ_y/\epsilon_0 &= j_y + \textstyle\sum_{snj} v_{snjy}, \\ \omega v_{snjz} &= c_{snj}v_{snjz} + b_{snj31}E_x + b_{snj32}E_y + b_{snj33}E_z, \\ \omega j_z &= b_{31}E_x + b_{32}E_y + b_{33}E_z, \\ iJ_z/\epsilon_0 &= j_z + \textstyle\sum_{snj} v_{snjz}, \\ \omega E_x &= c^2k_zB_y - iJ_x/\epsilon_0, \\ \omega E_y &= -c^2k_zB_x + c^2k_xB_z - iJ_y/\epsilon_0, \\ \omega E_z &= -c^2k_xB_y - iJ_z/\epsilon_0, \\ \omega B_x &= -k_zE_y, \\ \omega B_y &= k_zE_x - k_xE_z, \\ \omega B_z &= k_xE_y, \end{aligned}\right. \quad (10)$$

which yields a sparse matrix eigenvalue problem that can be readily solved using standard eigenvalue libraries. The symbols such as $v_{snjx}$, $j_{x,y,z}$, and $J_{x,y,z}$ do not have direct physical meanings but are analogous to the perturbed velocity and current density in the fluid-based derivations of plasma waves [25]. However, the elements of the eigenvector $(E_x, E_y, E_z, B_x, B_y, B_z)$ still represent the original perturbed electric and magnetic fields. Thus, the polarization of the solutions can also be obtained in a straightforward manner. The dimension of the matrix is $N_N = 3\times(N_{SNJ}+1)+6 = 3\times\{[S\times(2\times N+1)]\times J+1\}+6$, where $N$ is the number of harmonics retained for magnetized species, and $J$ is the order of the J-pole expansion for the $Z$ function. A surprising and unexpected feature of the above HH expansion is that the matrix structure and dimension are identical to the Maxwellian distribution case. Thus, the computation time remains the same, except for the time required to calculate the initial matrix coefficients. Other existing solvers for arbitrary distribution kinetic dispersion relations (e.g., [7, 8])

Same matrix dimension to the Maxwellian case, thus with similar computation time (except the time to calculate the initial matrix elements)!

Feature: Fast, accurate, arbitrary distributions, all solutions.

# 3. Obtain all solutions of kinetic dispersion relation

## 3.4 Arbitrary distributions (Xie2025)

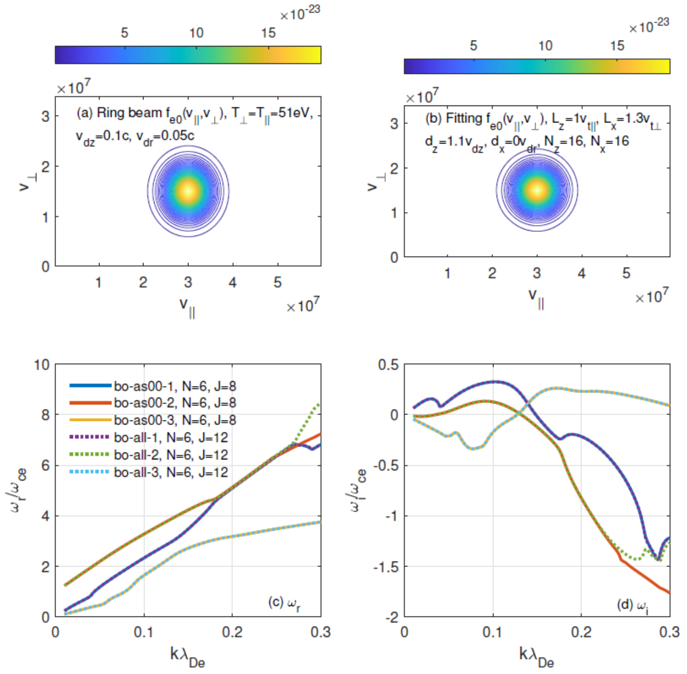


FIG. 2: Compare the unstable waves solutions under ring beam electron distribution for $\theta = 40°$, with analytical distribution and fitting distributions, where good agreement obtained except strong damped region. Here 'as00' means analytical distribution and 'all' means use Hermite expansion. Solve 120 wave vector points with all solutions, takes around 33s for analytical (N=6,J=8), 64s for fitting (N=6, J=8) and 200s for fitting (N=6, J=12).

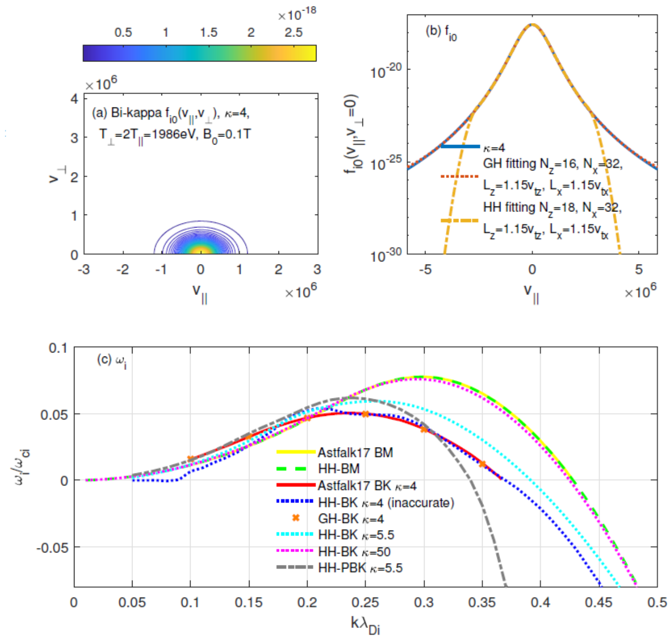


FIG. 4: Firehose instability under bi-Maxwellian (BM), bi-kappa (BK) and product bi-kappa (PBK) distributions, $\theta = 45°$.

Beyond other solvers: Irvine18, Verscharen18, Astfalk17, Matsuda92, Hellinger11, etc.

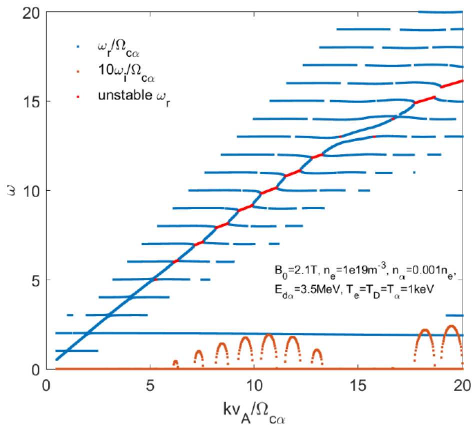


FIG. 3: Ion cyclotron emission for magnetic fusion plasma under drift ring ion beam distributions, $\theta = 89.5°$.

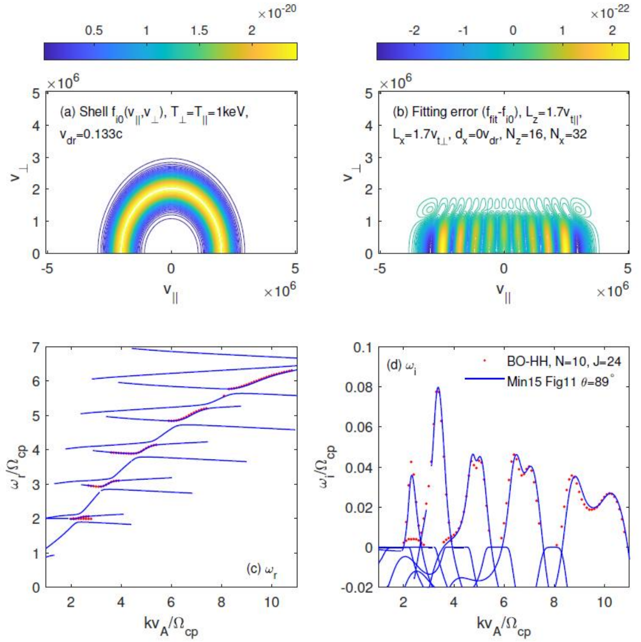


FIG. 5: Instabilities under shell ion distributions, $\theta = 89°$. The total CPU time for obtain all 100 wave vector points is around 70min for $S = 3$, $J = 24$ and $N = 10$.

- Test drift ring beam, kappa, shell distributions, all agree well.
- Support both instabilities and damped modes (except strong damped).

https://github.com/hsxie/boarbitrary

# 4. Summary and outlook

- An efficient framework for solving plasma waves with arbitrary distributions is developed, **which meets the requirements: fast, accurate, arbitrary distributions, all solutions, damped mode**.

- With combination of several key innovations: rational Pade J-pole approximation, expansion in basis functions, linear matrix transformation.

- Based on these series ideas, we have also developed a linear fluid plasma model with accurate kinetic Bernstein waves [*Phys. Plasmas* 32, 082110 (2025)].

- Future works: non-gyrotropic, relativistic, non-uniform, nonlinear, etc

All the above codes are available at github.
Suggestions are welcome. Thanks!

# **Appendix: Ray tracing code** (Xie2022CPC)

The ray tracing equations in Cartesian coordinates are

$$\frac{d\boldsymbol{r}}{dt} = \frac{\partial\omega}{\partial\boldsymbol{k}} = -\frac{\partial D/\partial\boldsymbol{k}}{\partial D/\partial\omega} = \boldsymbol{v}_g, \tag{1}$$

$$\frac{d\boldsymbol{k}}{dt} = -\frac{\partial\omega}{\partial\boldsymbol{r}} = \frac{\partial D/\partial\boldsymbol{r}}{\partial D/\partial\omega}, \tag{2}$$

with the dispersion relation

$$D(\omega, \boldsymbol{k}, \boldsymbol{r}) = 0, \tag{3}$$

where ray position $\boldsymbol{r} = (x, y, z)$ and wave vector $\boldsymbol{k} = (k_x, k_y, k_z)$. Here, $\omega$ is wave frequency, and $\boldsymbol{v}_g$ is wave group velocity. The geometrical optics approximation is valid in cases where the wave length is much smaller than the system nonuniform length, which is usually well satisfied for high frequency waves such as ECW and LHW, but should be used with caution for low frequency waves such as ICW and AW.

**Table A.2**
Fluid and kinetic plasma waves and instabilities code BO family [1–3].

| | Type | Names | References |
|---|---|---|---|
| BO family (open source) | dispersion relation ($\boldsymbol{k} \to \omega$) | PDRF, PDRK, BO, BO2.0 | Xie14, 16,19,21 |
| | ray tracing ($\omega \to \boldsymbol{k}$) | BORAY | Xie22 (this work) |

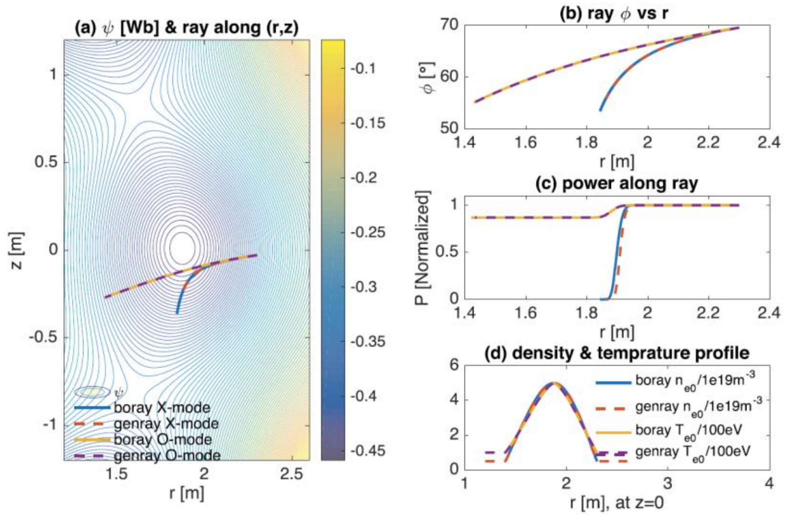


**Fig. 1.** Comparison of BORAY and GENRAY for EAST tokamak 100 GHz ECW O and X modes. Both ray trajectories and power absorptions agree well. (For interpretation of the colors in the figure(s), the reader is referred to the web version of this article.)

Support various axisymmetric configurations, tokamak/ST, FRC, mirror, dipole, etc

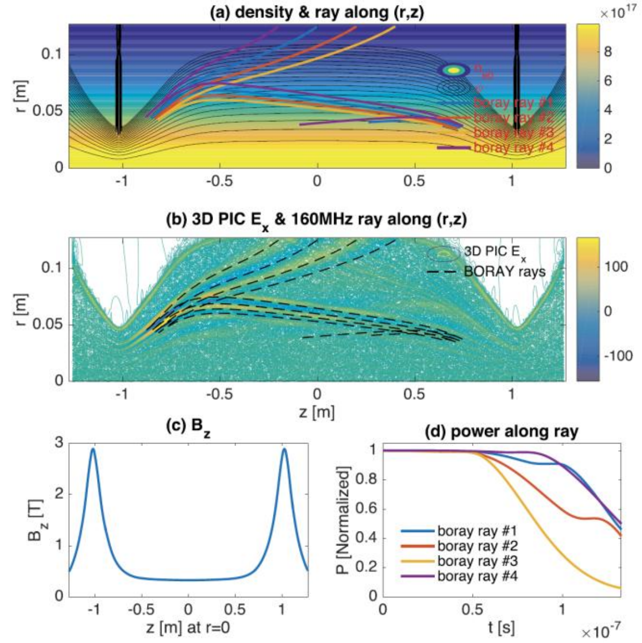


**Fig. 6.** Mirror 160 MHz LHW simulation results, which is close to the results in Ref. [17] of 3D PIC simulations results of the electric field, i.e., the ray trajectories from BORAY are just along where the electric field are strong in PIC results (b). The power absorptions (d) are also similar, i.e., the wave is almost decayed away before reaching the second turning point.

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https://github.com/hsxie/boray